ASTE-331a: Spacecraft Systems Engineering

Telecommunications (Telecom) and the End-to-End Information System (EEIS)

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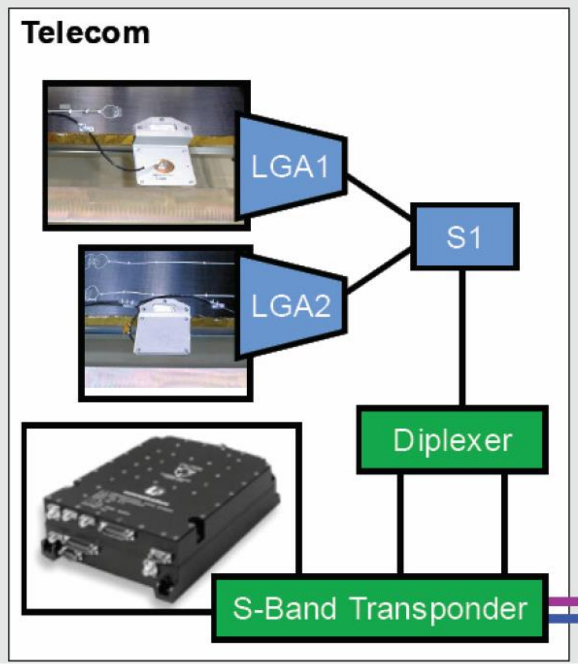
Telecom Overview (1 of 3)

- **Function**
  - Provides communication to/from the spacecraft. Typically with the Earth, but can include other spacecraft.
    - Command reception and detection
    - Telemetry modulation & transmission
    - Bandwidth selection & antenna pointing
    - Carrier tracking modes and ranging
  - Occasionally used for radio science
  - Primary driver is typically spacecraft-Earth distance

- **GRAIL Example**
  - S-band transponder
  - Low-gain antenna (LGA)
  - Misc. hardware
  - Total = ~5 kg



- **Common Components**
  - Radio/transponder
  - Amplifier
  - Antennas
    - High-gain (HGA)
    - Medium-gain (MGA)
    - Low-gain (LGA)

- **Key Trades & Analyses**
  - Fixed vs. articulated antennas
  - Data rate & volume analysis
  - Link budget & antenna sizing
  - Radio-band and power level
  - Heritage from prior systems

- **Key Parameters**
  - Mass, power, and cost
  - Transmission power, data, & margin

Telecom Overview (2 of 3)

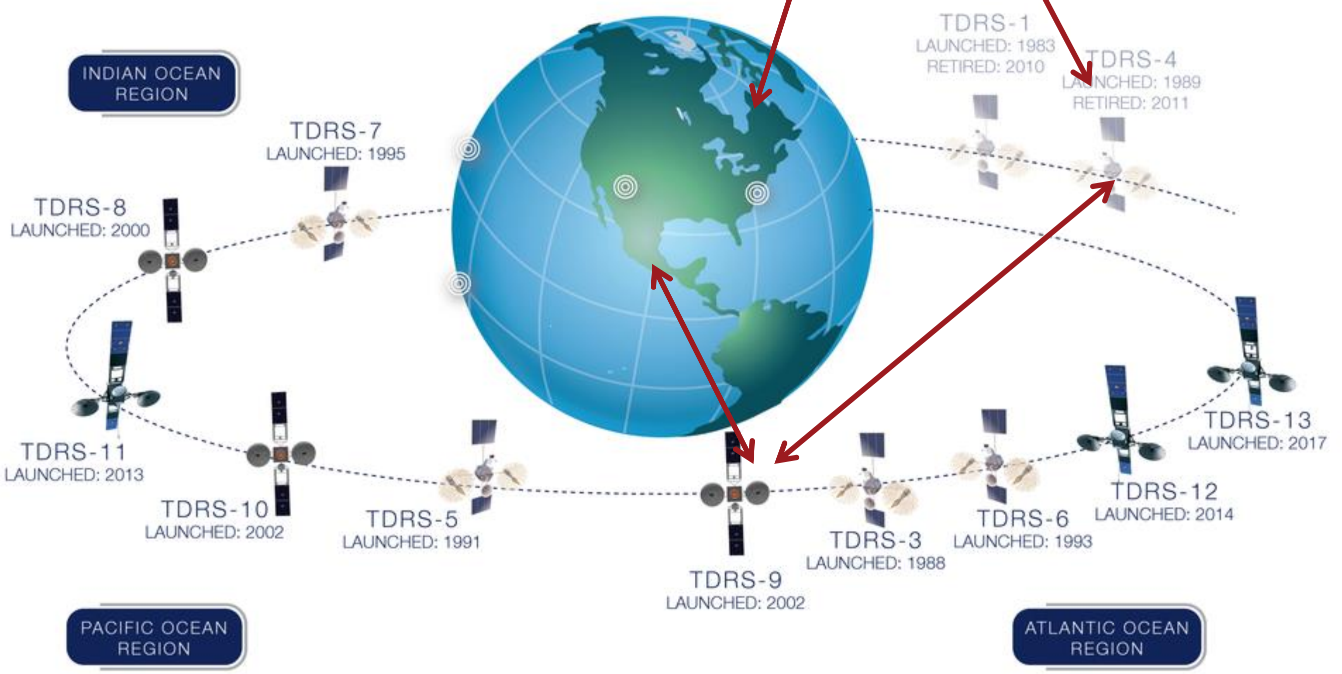
- **Communications is critical for nearly every spacecraft**
  - Most likely subsystem to be redundant
  - Direct uplink/downlink comm. with specific ground stations
  - Relay communication, such as using the TDRS network or via MRO (from Mars)
  - Significant increases in communication over the past 30 years (≥ 1 Gbps possible)

Space Vehicles that use TDRS
- ISS, Hubble, NASA LEOs, etc.
- Most can also downlink directly…

Direct Communications

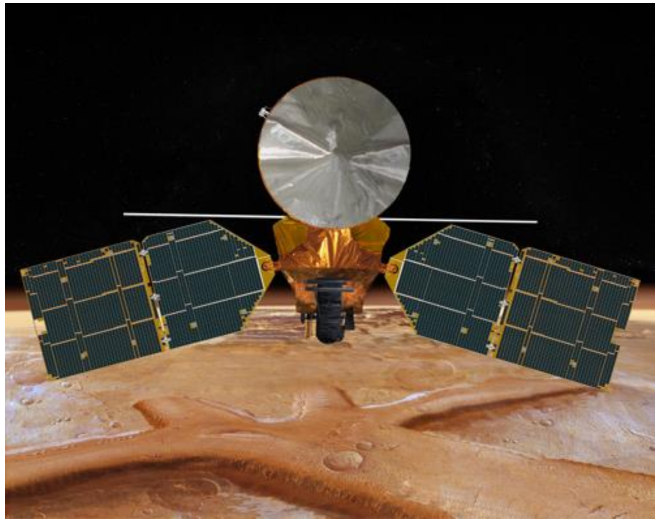
TDRS Relay Communications

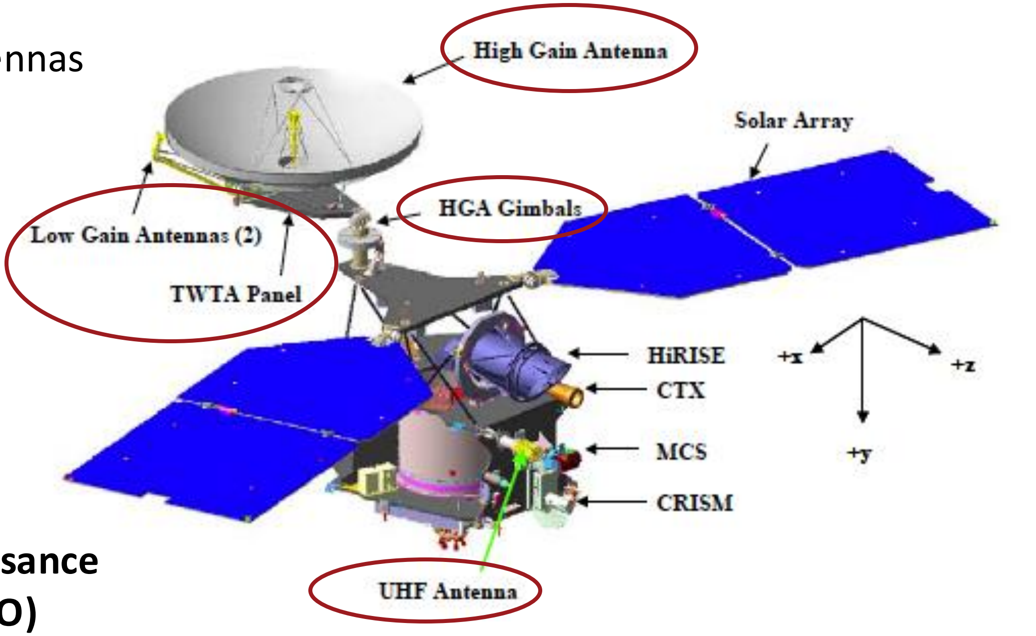
Tracking & Data Relay Satellite (TDRS)



Telecom Overview (3 of 3)

- **Functions**
  - Command reception and detection
    - Acquire, track, detect uplink data
    - Forward command data to avionics
  - Telemetry modulation & transmission
    - Receive telemetry data from avionics
    - Modulate downlink signal (synchronize, packetize, encoding, encryption, etc.)
  - Perform antenna pointing
  - Select desired antenna (or band)
    - In safe-mode, spacecraft will typically default to low gain antennas
  - Carrier tracking modes
  - Ranging





Mars Reconnaissance Orbiter (MRO)

Telecom Hardware Overview

[eg, C&DH]

Digital Signal

[eg, SDST]

RF Signal

[eg, TWTA]

DPXR

WTS

Avionics Subsystem

The avionics subsystem serves as the interface to the telecom system. This often consists of an "ULDL" card that decodes and relays the uplink data to FSW and/or directly to hardware devices.

Radio/Transponder

An electronics box responsible for:
Uplink: Receiving the formatted bits from C&DH, converting them to an RF signal, and transmitting to the antennas.
Downlink: Receiving the RF signal from the antennas, converting them to bits, and relaying to the C&DH system.

Amplifier

An RF power amplifier converts a low-power RF signal to a higher power RF signal that drives the spacecraft antennas.

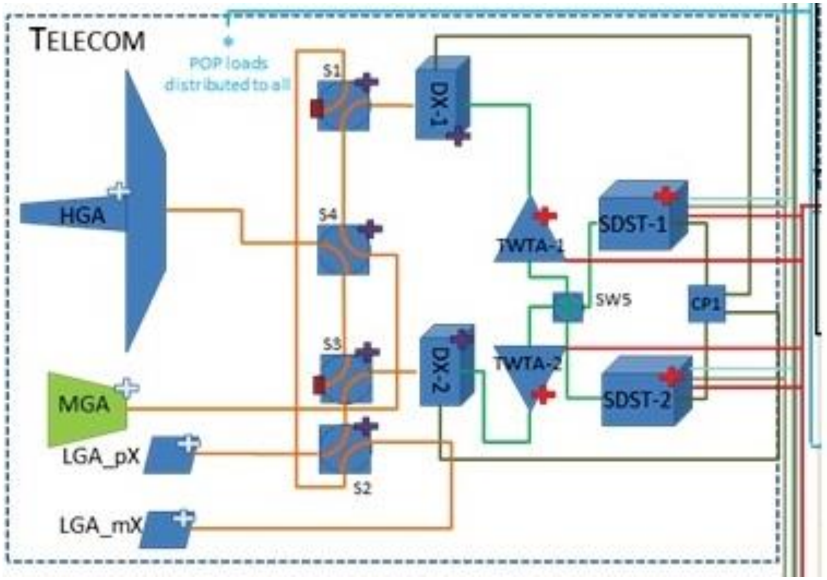
Diplexer(s)

Passive devices that combine and/or separate uplink/downlink signals.

Waveguide Transfer Switch(es)

Switches that route an RF signal to a selected device/antenna.

OSIRIS-REx Telecom Block Diagram



Low Gain Antenna (LGA)

Typically, multiple dipole antennas that ensure ~360-deg coverage at a low data rate that is adequate for critical uplink/downlink.

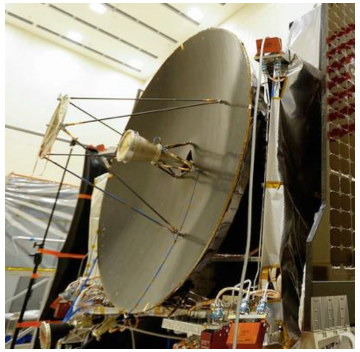
Medium Gain Antenna (MGA)

When used, this is typically a horn antenna that provides higher data rates than an LGA, but less rigorous pointing requirements than an HGA.

High Gain Antenna (HGA)

Typically, a single parabolic dish that provides high data rate, but requires narrow pointing control.

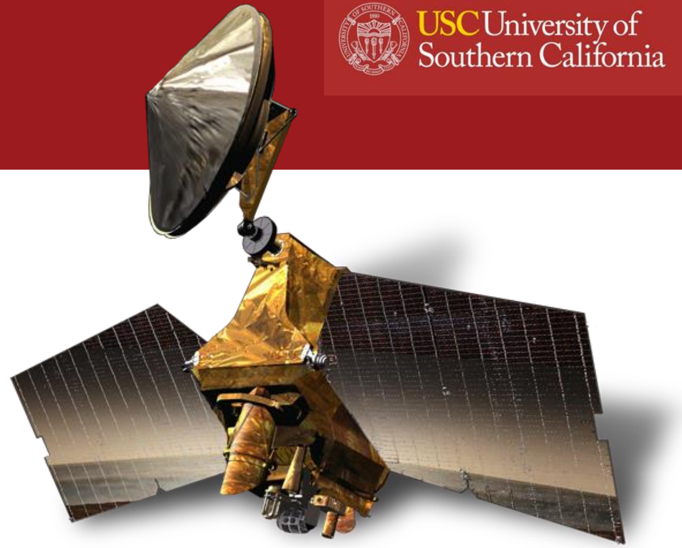
OSIRIS-REx 2.1 m HGA



What component in the diagram is not described?
Speculate on its use and when/why it's used.

Lesson Learned (#1796)

MRO Waveguide Transfer Switch (WTS) Anomaly



Mission

- Mars Reconnaissance Orbiter (MRO) is a spacecraft designed to study the geology and climate of Mars, provide reconnaissance of future landing sites, and relay data from surface missions back to Earth. It was launched in August of 2005 and arrived in March of 2006.
- The spacecraft continues to operate far beyond its intended design life. Due to its critical role as a high-speed data-relay for ground missions, NASA intends to continue the mission as long as possible, at least through the late 2020s.

Driving Event & Root Cause

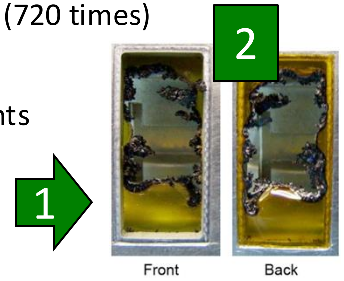
- For the downlink, MRO uses a WTS to route a 100-W RF signal from one of the two power amplifiers to one of two radiating antennas.
- Five months after arrival, the WTS failed to actuate. The onboard software maintained the commanded downlink configuration by commanding a switch to the redundant X band amplifier. Telemetry indicates that the switch is stuck between its two nominal positions, causing the switch rotor to partially block the RF energy passing through the switch. This has resulted in a downlink RF power loss (~1 dB) and a temperature increase (~15 deg C) caused by absorption and dissipation of the reflected energy.
- The most likely root cause is the following series of events:
  1. Accumulation of conductive debris (flaked plating?) on the polyimide tape windows used for WTS venting
  2. RF breakdown of these surfaces resulting in high temperature decomposition of the tape
  3. Large amount of debris from tape entering the WTS and causing the switch to bind
  4. Reduced performance of RF signal by 1 dB (accommodated by 3 dB of margin)
- This process was exacerbated by the frequency of MRO's use of the switch (720 times)

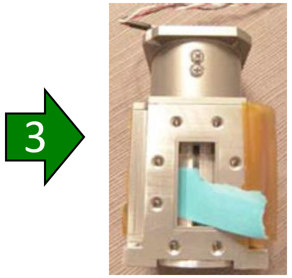
Lessons Learned

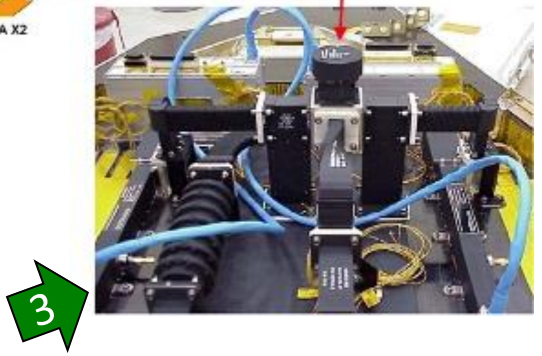
- Avoid using polyimide tape windows for contamination control on WTS vents
- Many additional recommendations on analysis & design

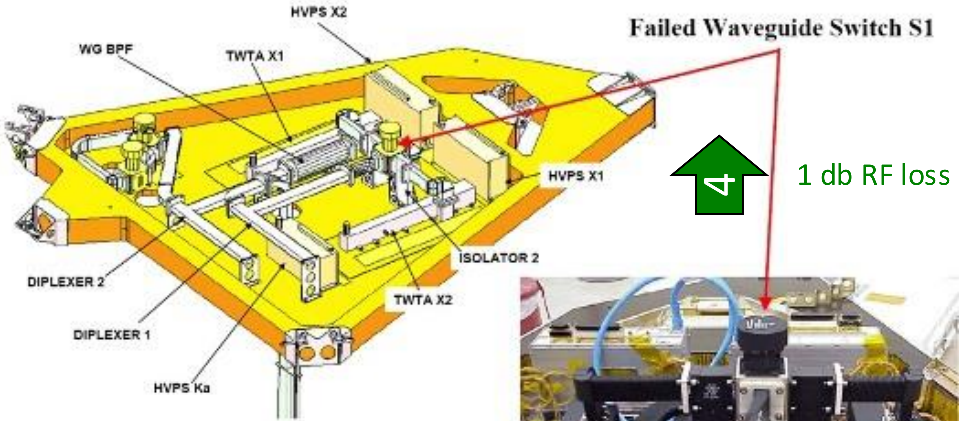
Source: JPL/NASA Lessons Learned

- https://llis.nasa.gov/lesson/1796



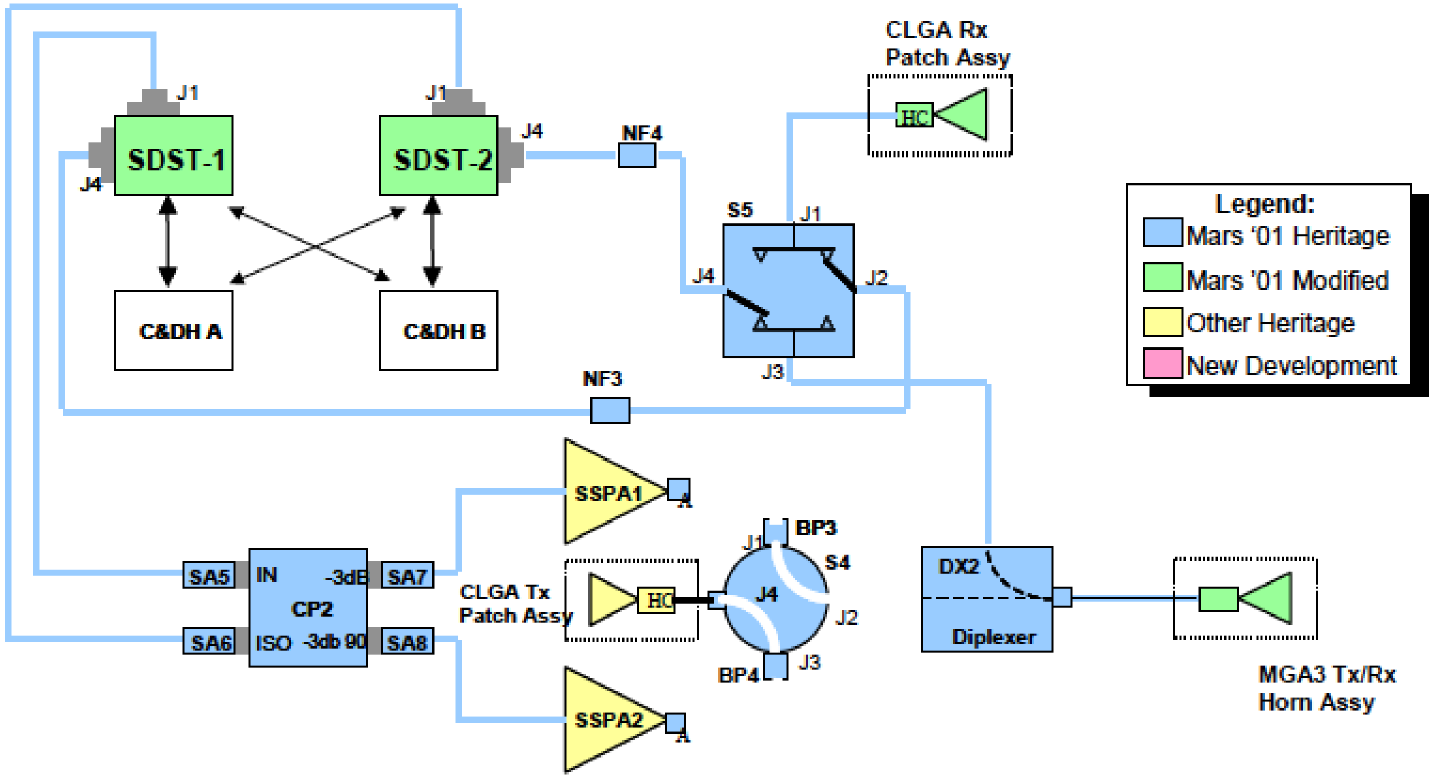






Tape & WTS pictures show testing that was completed as part of the anomaly investigation

Phoenix Telecom Subsystem (X-Band)



Source: “Phoenix Telecommunications”, DESCANSO Series, 2010

What component in the diagram is not described?
Speculate on its use and when/why it’s used.

Mission Description

- NASA science mission to investigate the geologic history and biological potential of Martian arctic
- Launch in 2007, arrived May 2008, ended Nov. 2008

Telecom System

- 2-way X-band for communication & ranging in Cruise
  - 2 SDST radios and 2 15-W SSPA amplifiers
  - 1 medium gain (MGA)
  - Separate low gain antennas (Rx & Tx)
  - 2100 bps (Earth) to 40 bps (Mars)
- 2-way UHF radio for EDL & Surface
  - 1 helix antenna (HGA) and 1 monopole (LGA)
  - 32 kbps (EDL) to 256 kbps (Surface)

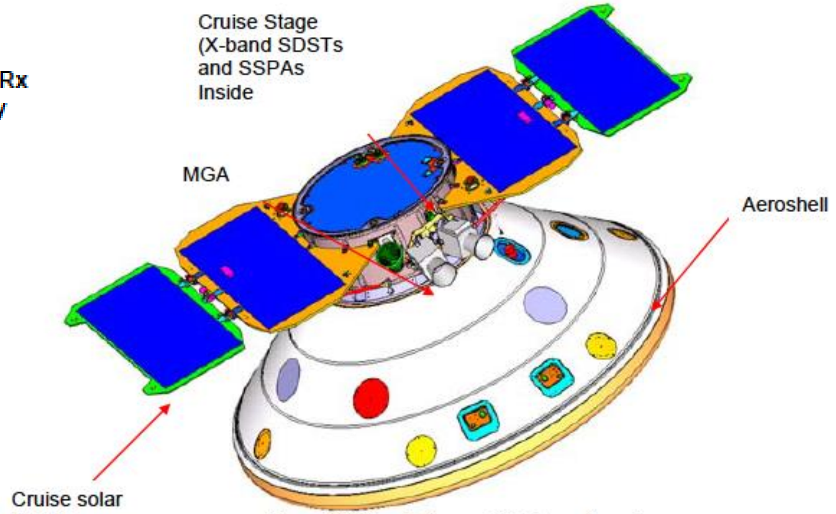
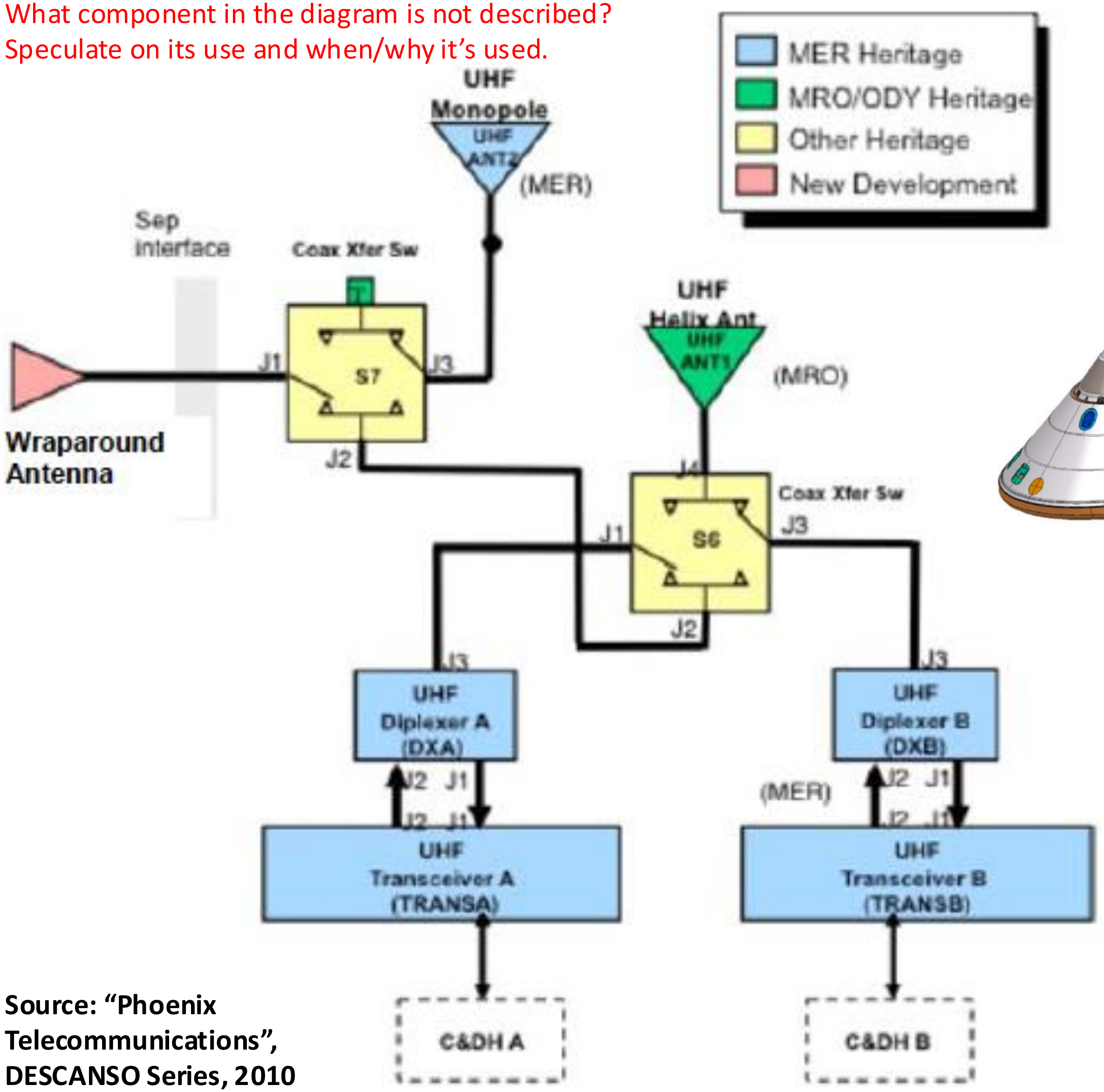


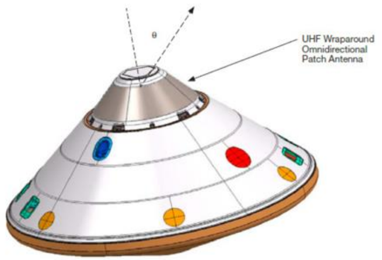
Figure 2-14 Phoenix Spacecraft Cruise configuration.

Phoenix Telecom Subsystem (UHF)

What component in the diagram is not described?
Speculate on its use and when/why it's used.



Source: "Phoenix Telecommunications", DESCANSO Series, 2010



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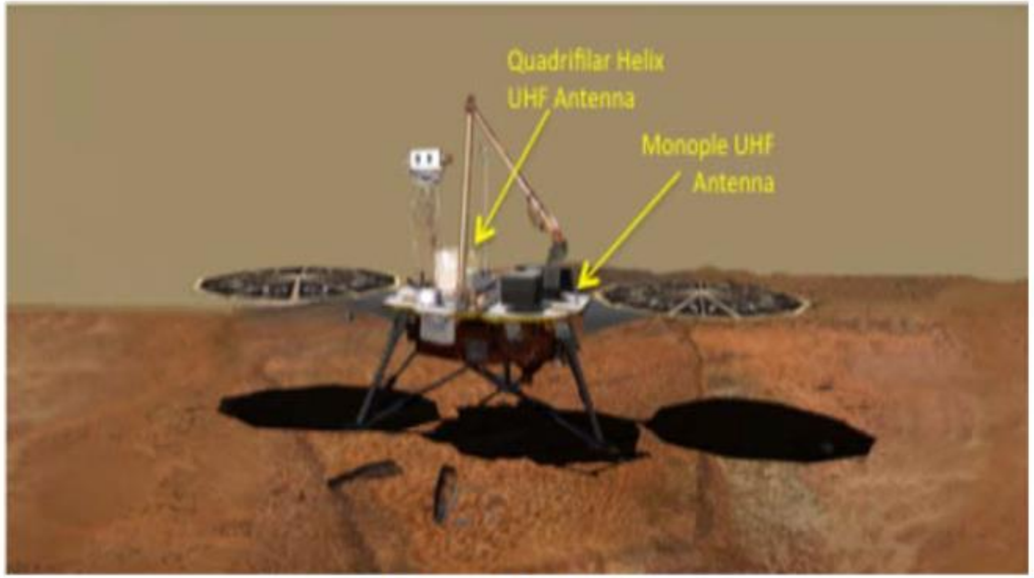
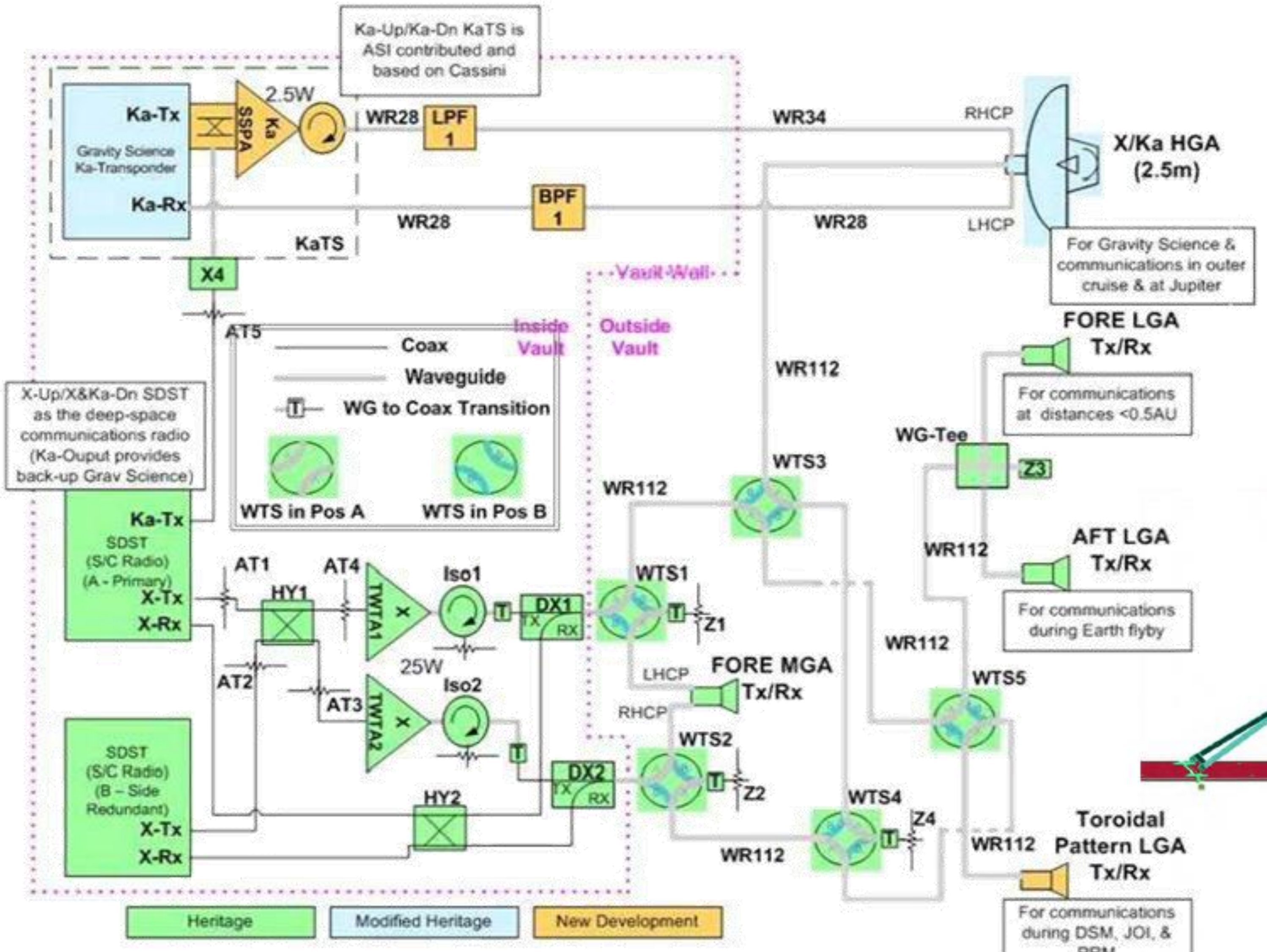


Juno Telecom Subsystem



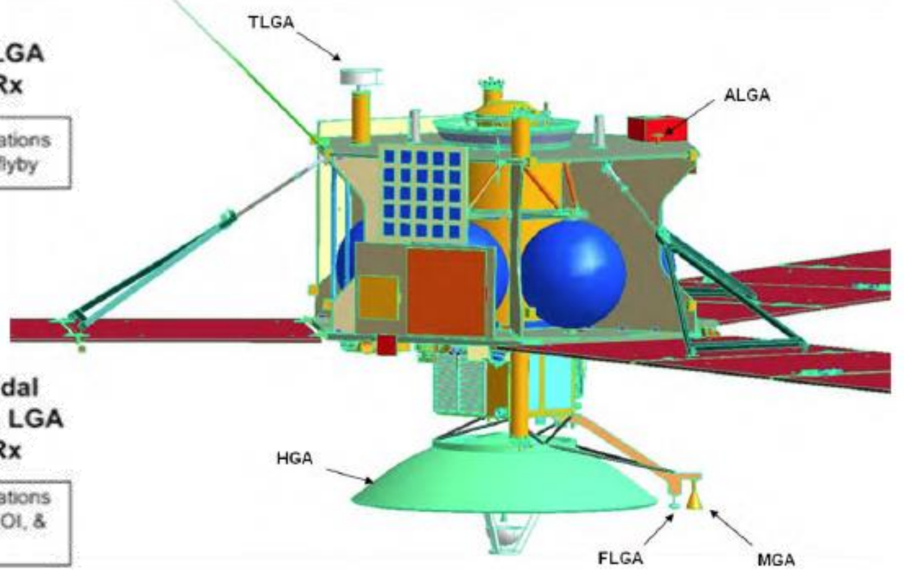
Source: “Juno Telecommunications”, DESCANSO Series, 2012

Mission Description

- NASA science mission to understand the formation and evolution of Jupiter
- Launched in 2011, arrived 2016, extended to 2025

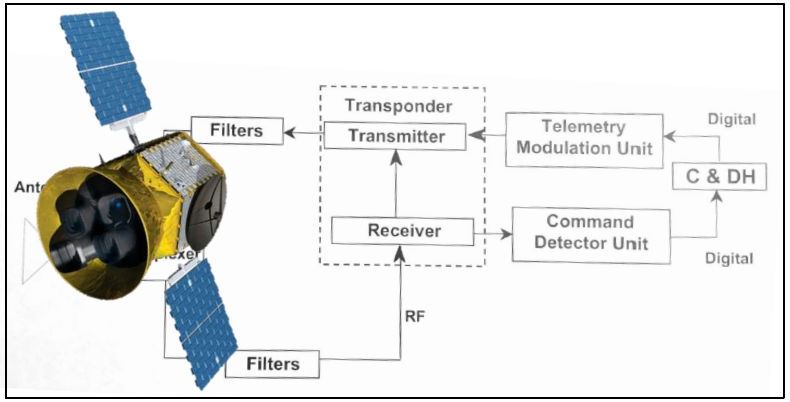
Telecom System

- 2-way X-band for communication & ranging
  - 2 SDST radios and 2 TWTA amplifiers
  - 1,745 bps (launch), 100 bps (cruise), 18 kbps (HGA at Jupiter), 10 bps (safe)
- 2-way Ka-band for gravity science (single-string)
  - 1 Italian radio and 1 SSPA amplifier
- Five Antennas
  - 2.5 m HGA, MGA, and 3 LGAs



End-to-End Information System (EEIS) Overview

- **EEIS stretches beyond the telecom subsystem to encompass all functions, data formats, and path losses from the initial selection and formatting of the data for transmission to the final reception and use of the data**
  - These are the to/from paths between telecom and ground (uplink & downlink)
- **Flight-Ground Interface Control Document (FG-ICD)**
  - This document describes the interfaces & data formats at each step in the process
  - To maximize efficiency, data is transmitted in binary and thus needs to be converted at several steps along the full path
- **Link Budget**
  - This is the corresponding analysis that captures all of the losses (L) and gains (G) between the source & receiver
  - This budget determines the data rate (in bits per second, bps) at a given bit error rate (BER)
  - It’s used across many industries to design communication systems



Telecom Subsystem

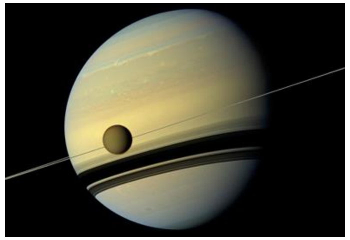
Space Loss

Atmospheric Attenuation

Ground Station

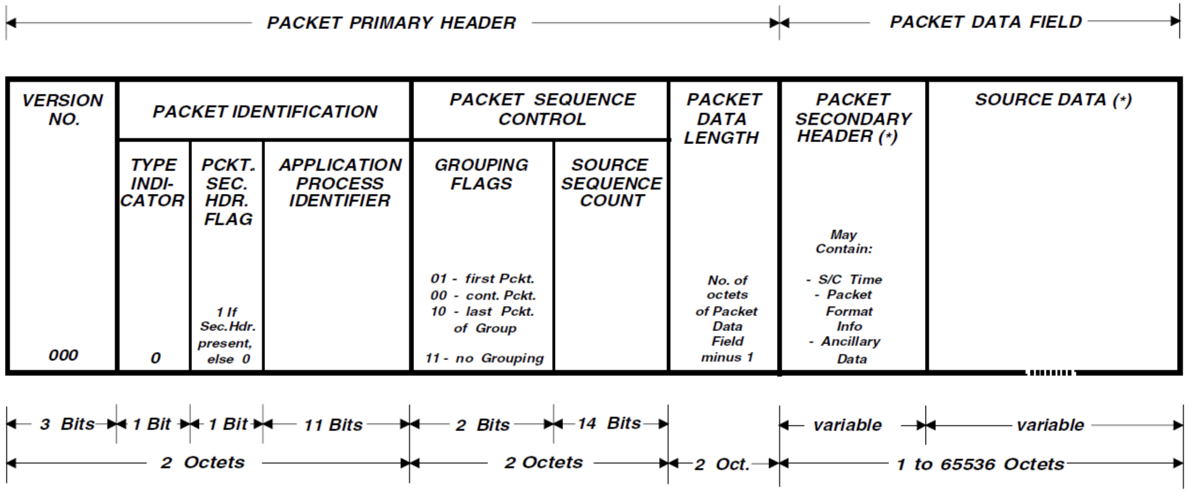
End-to-End Information System (in bits)

Picture on Titan



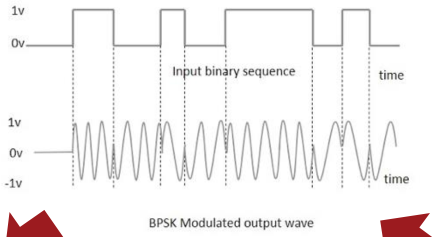
Binary
01000000
01000001
01000010
01000011
01000100
01000101
01000110
01000111
01001000
01001001
01001010
01001011
01001100

Flight Ground Interface Control Document (FG-ICD)

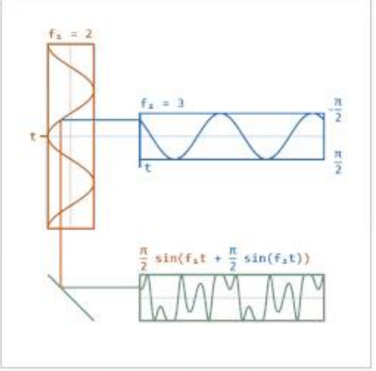


Picture on Earth

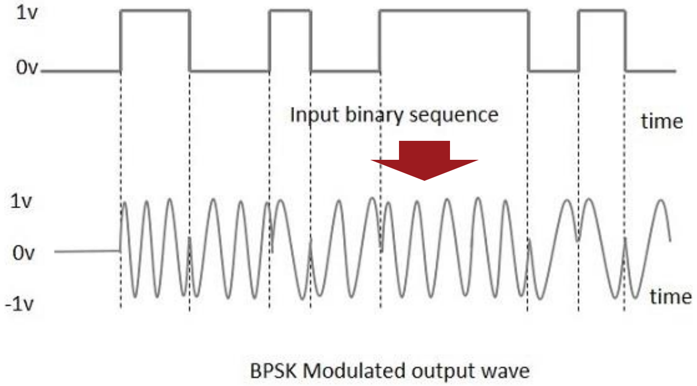
ASCII	Hex	Binary
@	40	01000000
A	41	01000001
B	42	01000010
C	43	01000011
D	44	01000100
E	45	01000101
F	46	01000110
G	47	01000111
H	48	01001000
I	49	01001001
J	4A	01001010
K	4B	01001011
L	4C	01001100



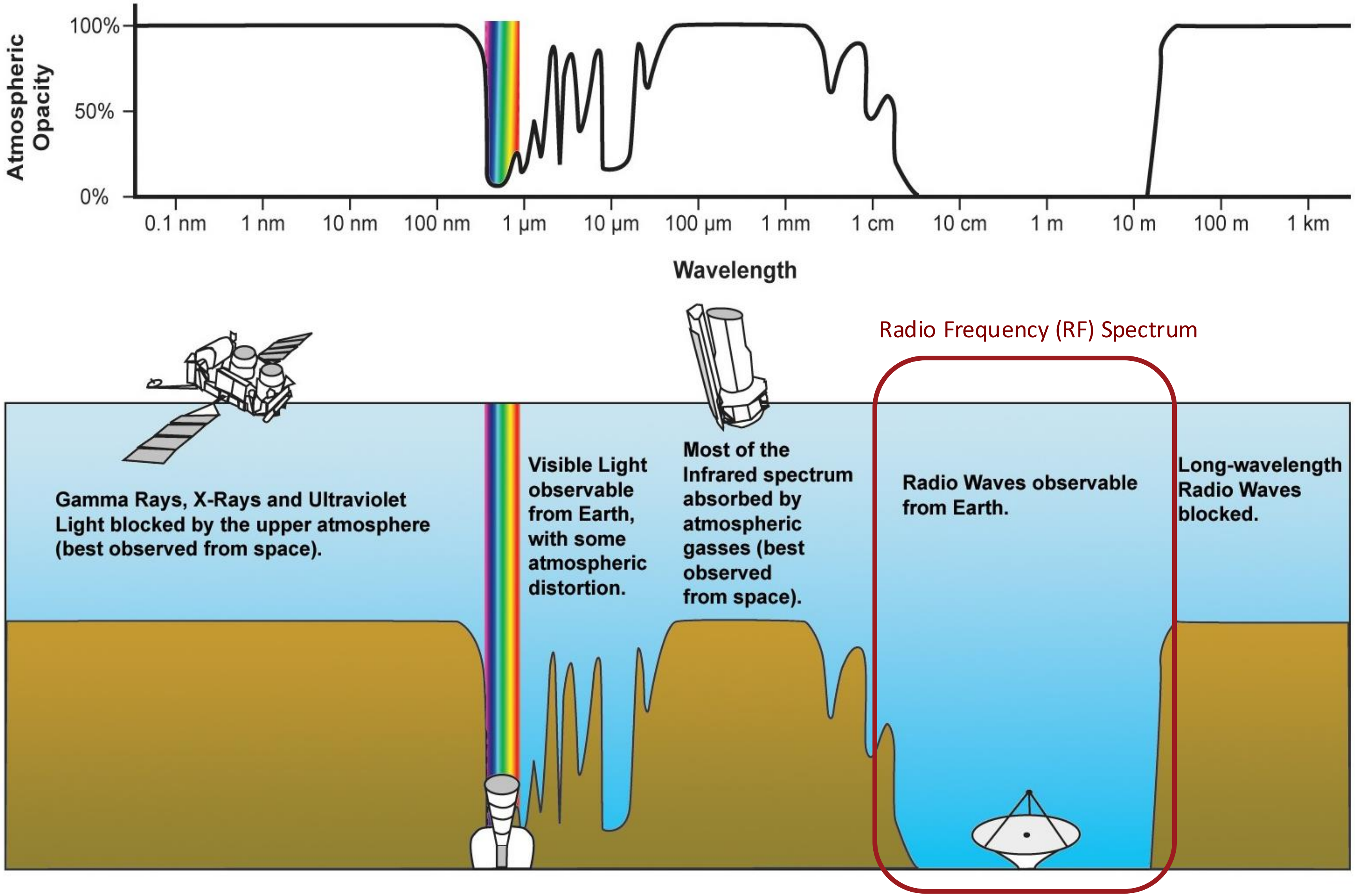
(relay via Cassini s/c)



The modulating wave (blue) is modulating the carrier wave (red), resulting the PM signal (green). g(t) = π/2 * sin(2*2πt+π/2*sin(3*2πt))



Electromagnetic (EM) Spectrum

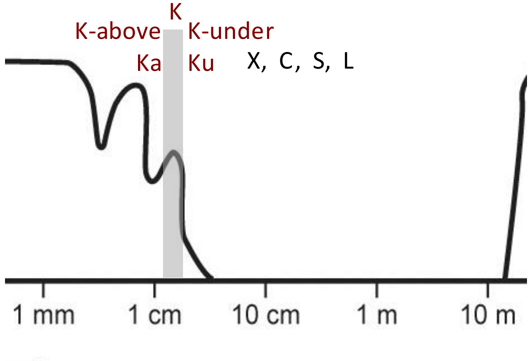


The EM spectrum describes light in the form of frequencies and their respective wavelengths and photon energies.

The EM spectrum is used heavily in space, where its used or mitigated in the following applications:

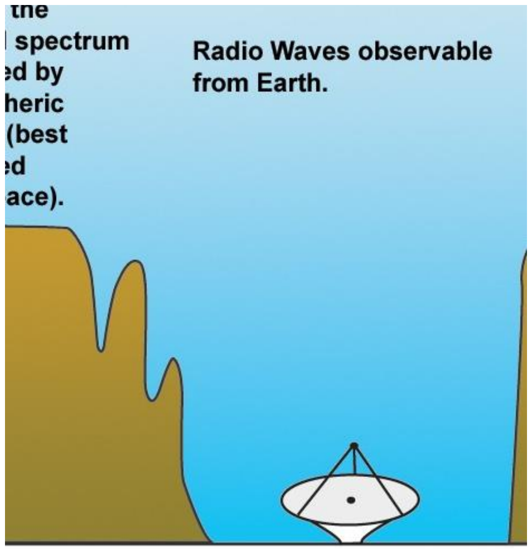
- Radiation shielding
- Power generation (eg, solar arrays)
- Thermal absorption & radiation
- Science (eg, imaging, spectroscopy, radar)
- Communications

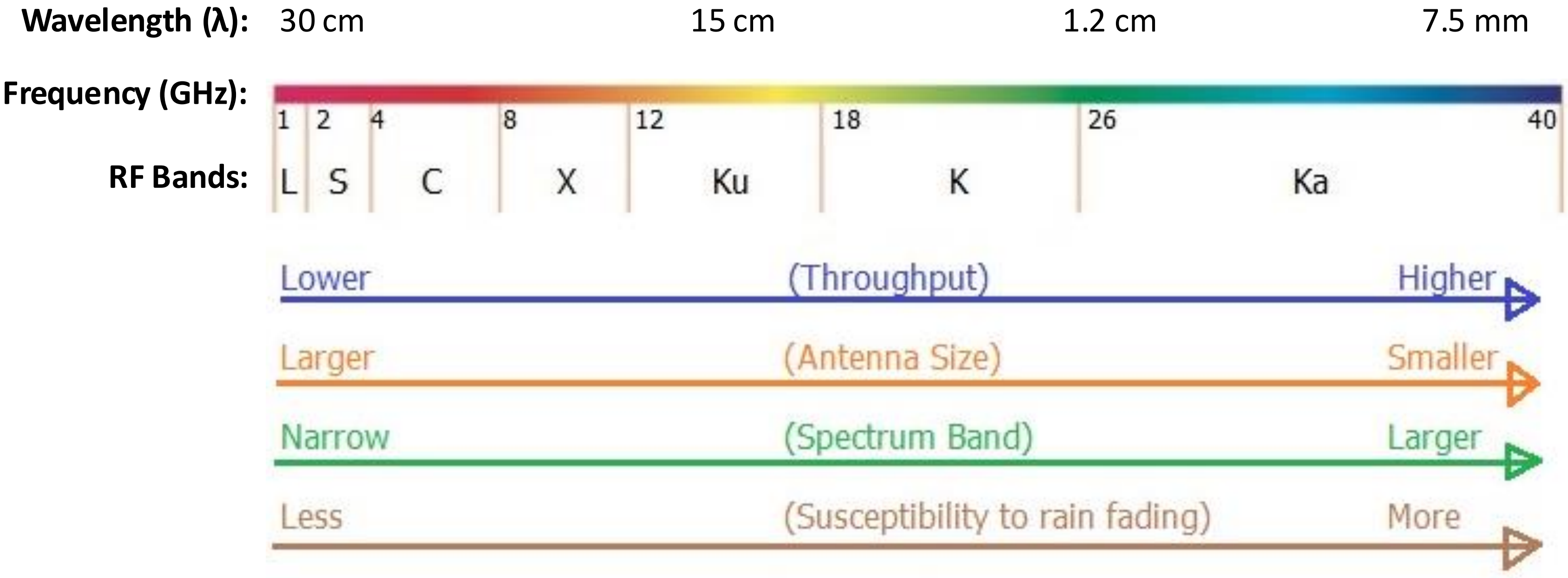
Radio Frequency (RF) Spectrum



Key Relationship: $f = c / \lambda$

Where: f is frequency in Hz (cycles/sec)
c is the speed of light (m/s)
λ is wavelength (m)

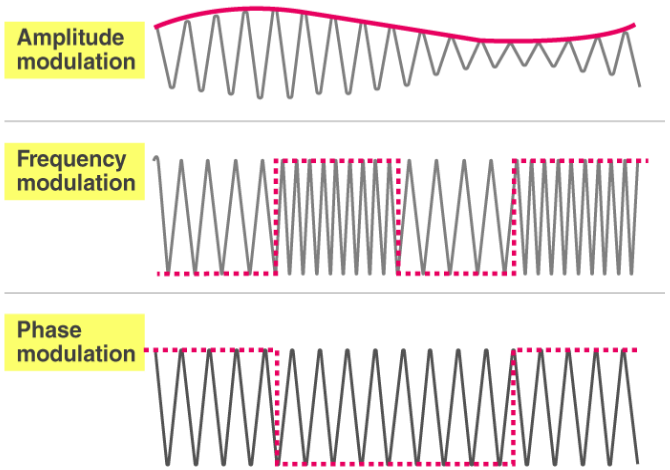




	Advantages	Disadvantages	Notes
Lower in Spectrum L, S, C, X (robust)	• Frequently used with small omni-antennas • Less susceptible to atmospheric effects (eg, rain, fog, snow)	• More power required • Lower throughput • Larger HGA size	• NASA/DoD use S- & X-bands • Used for low or high rate data • GPS, comm. satellites, & mobile phones use L-band • TV broadcasting uses C-band
Higher in Spectrum Ku, K, Ka (high-performance)	• Lower power required • Higher throughput • Smaller HGA size	• More susceptible to atmospheric effects (eg, rain)	• Communications/NASA use Ka-band • Used for high rate-data • Satellite comm. & direct broadcast use Ku-band • K-band avoided due to atmosphere

A Few Definitions…

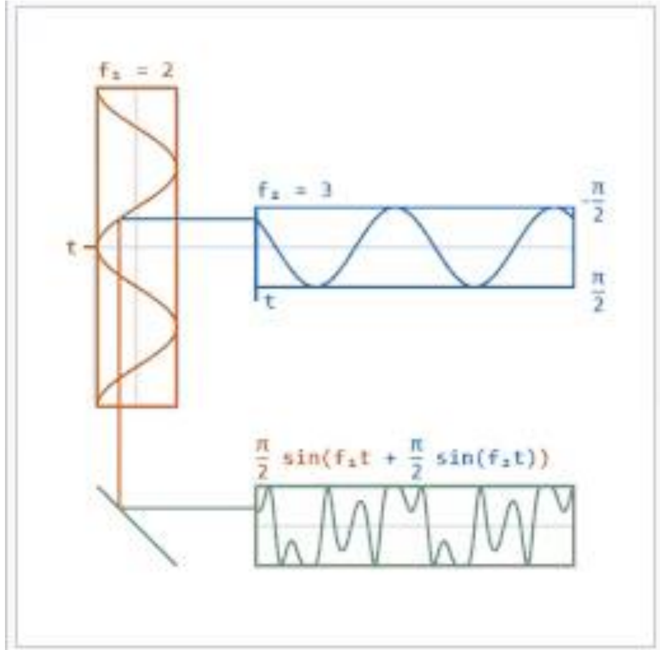
- **Uplink, Downlink, Crosslink**
  - Communication up/down from the ground across via relay to other spacecraft
- **Carrier**
  - Radio frequency (RF) source used to carry information
  - Frequency: $f = c / \lambda$ (typically in units of Hz; that is, cycles/second)
- **Modulation**
  - The process by which an input signal varies the characteristics of a carrier wave, including the amplitude, phase or frequency
  - Spacecraft downlinks typically use phase modulation, where the amplitude/frequency remain the same and the phase (ie, signal level) is changed
- **Demodulation**
  - The process for restoring the original input signal
- **Bit Error Rate (BER)**
  - Frequency/probability of a bit error rate
  - Typical requirement is 10^-5 (ie, 1 in error in 100,000 bits)



Phase modulation varies the signal level (very slightly)

Carrier
(carries your data)

Modulating Wave
(injects your data)



The modulating wave (blue) is modulating the carrier wave (red), resulting the PM signal (green). $g(t) = \pi/2 * \sin(2*2\pi t + \pi/2*\sin(3*2\pi t))$

Isotropic & Anisotropic Radiation

Surface area of a sphere = $4 \pi r^2$

Therefore, assuming no losses due to inefficiency, a 1 W source will gradually dissipate with $1/r^2$.

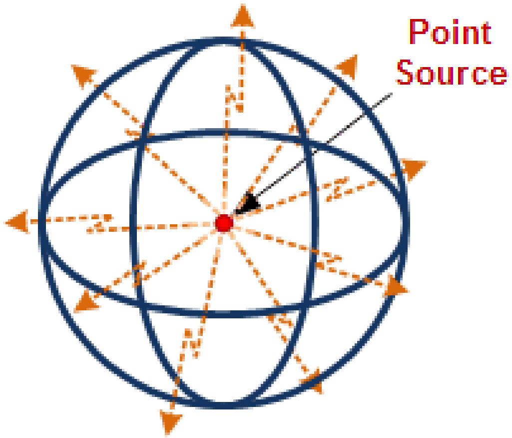
Or, flux density (in Watts) = $P_t \; / \; (4 \pi r^2)$

If a transmitter has 'gain', then it transmits a signal in a given direction at an increased level of power than if it were isotropic. That is:

Gain = $A_{effective} \; / \; A_{isotropic}$

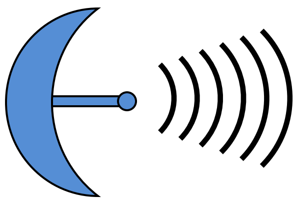
…for a given section of the surface area (A).

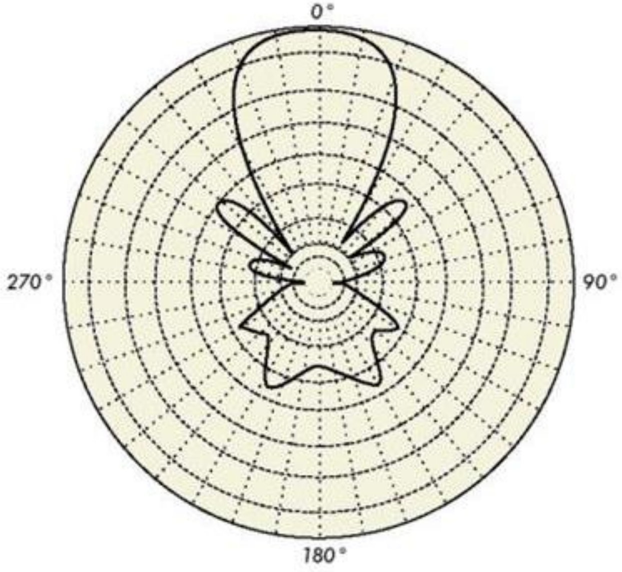
Isotropic Radiation



Equal power radiated in all directions from the center of sphere with a point source (ie, equipotential electric field at all constant radii).

Anisotropic Radiation





Radiation where its intensity depends on the direction. Typically described via an antenna gain pattern.

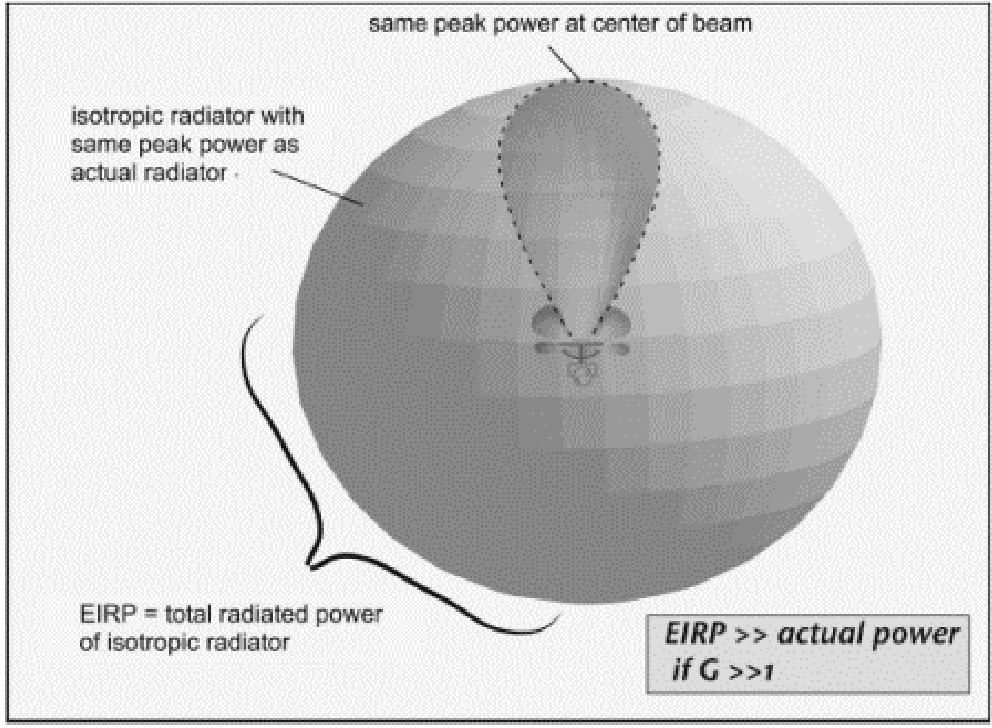
Effective Isotropic Radiated power (EIRP)

- **Effective Isotropic Radiated Power (EIRP) is the resulting transmitted power after all gains & losses during transmission have been factored in, including:**
  - Antenna gains ($G_{transmitted}$)
  - Antenna pointing and/or circuit losses ($L_{antenna}$, $L_{circuit}$)

- **Thus EIRP = P_t G_t L_a Lc**
  - Where EIRP & Pt are in Watts and G & L are unitless factors
  - In other words, the resulting power is the original power times a series of factors (gains & losses)

- **Unfortunately, for communications, gains & losses typically vary by several orders of magnitude, such as:**
  - Gains: x 10-1,000,0000
  - Losses x 1/10th to 1/1,000,000th

- **The result is that it's far easier to work in decibels (dB)**



Additional Equations

Radiated Power

- This equation converts transmitted power (P_t) to radiated power (P_r)

$$P_r = \frac{A_r}{4\pi R^2} P_t L_a$$

Where…

R = Range
Ar = Receiving area
Pr = Power received
Pt = Power transmitted
La = Transmission loss factor
G = Transmission

Telecom Link Equation

- E_b/N_0 is the final result that determines performance. E_b is energy per bit, N_0 is noise power spectral density in Watts per hertz
- $E_b/N_0 = P_t L_t G_t \left(\frac{\lambda^2}{4\pi R_g^{\,2}} \right) L_m G_r L_r \left(\frac{1}{R_t} \right) / N_0$
- Where:

P_t = transmit power
L_t = transmit circuit loss
G_t = transmit antenna gain
lambda = wavelength
L_m = medium loss
R_g = range
G_r = receive gain
L_r = receive circuit loss
R_t = bit rate

The decibel (dB)

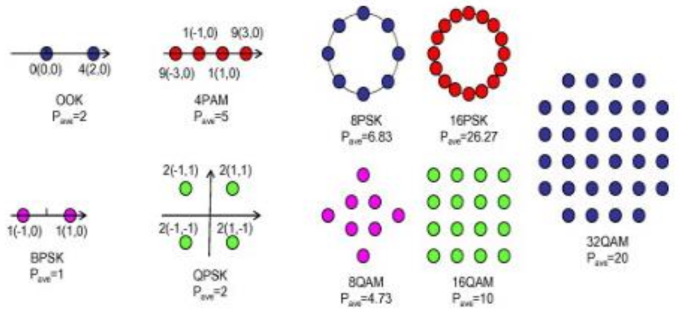
- **Background**
  - Consider dB as just a method that is used to express ratios (ie, gains & losses).
    - dB converts factors (gain or loss) using a Log10 function; that is, $10 \times \text{Log}_{10} (G) = G$ (in dB)
    - The inverse of this is: Gain = $10^{(\text{Gain (in dB)} / 10)}$
  - Examples:
    - Example 1: 10 W increases by a factor of 2, 10 x 2 = 20 W
      - Factor of 2 = $10 \times \text{Log}_{10} (2)$ = 3 dB
    - Example 2: 10 W decreases by a factor of 2, 10 / 2 = 5 W
      - Factor of ½ = $10 \times \text{Log}_{10}$ (½) = $10 \times \text{Log}_{10} (1) - 10 \times \text{Log}_{10} (2)$ = 0 dB – 3 dB = -3dB
  - This is important because of the properties of log functions
    - Example 3: 10 W x 1/100 x 1,000 x 1/500 x 80,000 x 1/20,000 = 0.8 W
      - Where as: 10 W x (-20 dB + 30 dB – 27 dB + 49 dB – 43 dB) = 0.8 W → In other words, we can just add/subtract gains & losses by using dB
        10 W x (-11 dB) = 0.8 W
        10 W x 0.08 = 0.8 W, where 0.08 = 10^(-11/10)

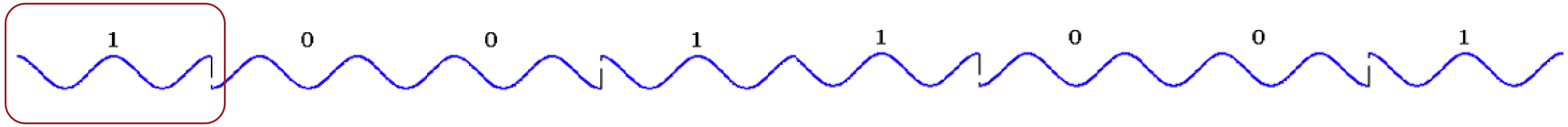
- **Two additional conventions**
  - Watts can be converted to dBi: $10 \times \text{Log}_{10}$ (Power in W) = Power in dBi
  - You can convert to milliWatts as well (in dBm): $10 \times \text{Log}_{10}$ (Power in W x 1000) =
    10 x Log10 (Power in W) + 10 x Log10 (1000) =
    Power in dBi + 30 = Power in dBm

Questions from Prior Class

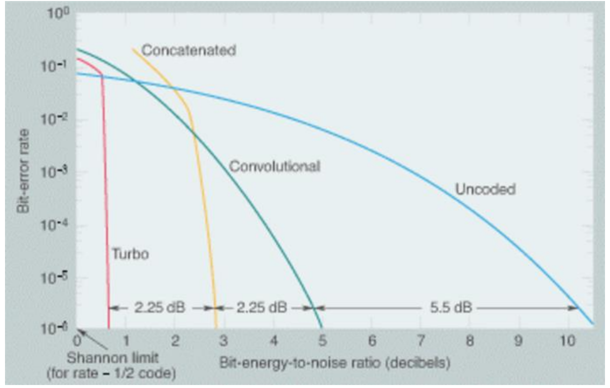
- **How much power devoted to telecom?**
  - Transmitters are typically in the 10 to 100 W class
  - Overall power is often 100 to 1,000 W, so 10-20% is probably a good assumption
- **In telecommunication, what is a symbol?**
  - A symbol is a repeated wave pattern that represents a specific # of bits (eg, 1 bit in BPSK)

Phase Key Shifting

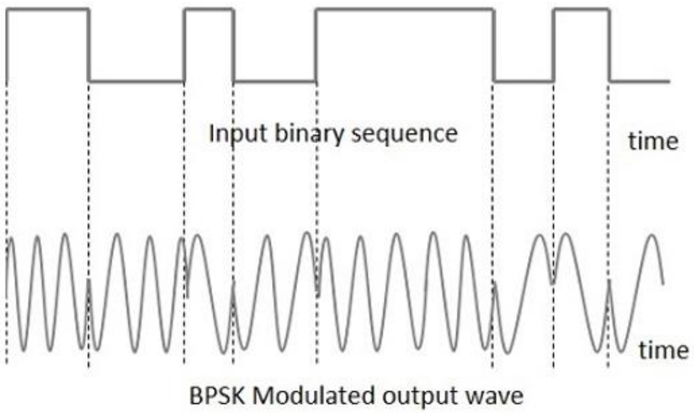




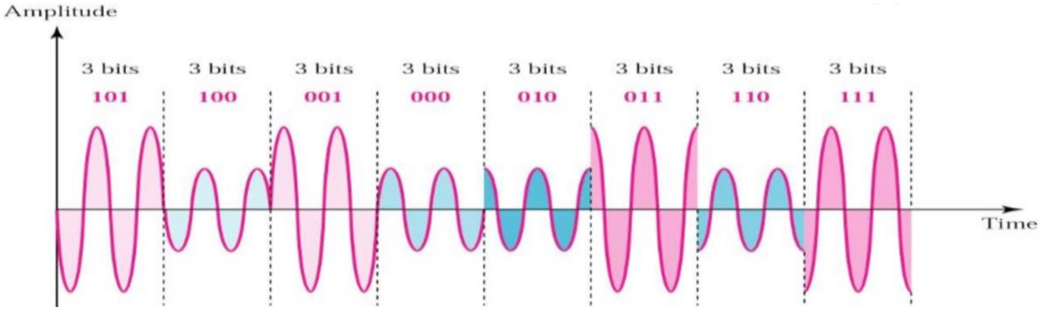
Error Correction Codes



BPSK (1 bit/symbol)



8-PSK (3 bits/symbol)



Link Budget Overview

Total Received Power			
Variables	Units	Values (70-m)	Values (34-m)
Total Received Power	db	-118.1	-124.3

Resulting Downlink Rate			
Variables	Units	Values (70-m)	Values (34-m)
Power Transmission over Noise	dB	65.9	59.7
Noise Floor	dB	-184	-184
Encoding & Telemetry Support	dB	0.4	0.4
Margin	dB	4	4
Downlink Rate	dB	61.5	64.1
Downlink Rate	bps	1,416,677	2,577,928
Downlink Rate	Mbps	1.417	2.578

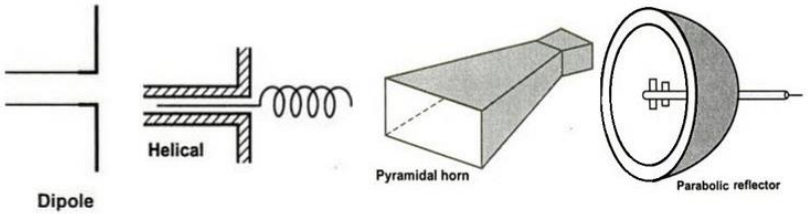
Telecom & Transmitter Power Calculations		
Variables	Units	Values
Radio/TWTA Power Input	W	100.00
Bandwidth	MHz	8400
Radio Efficiency	-	0.37
Transmitter Power	W	37.29
Transmitter Power	dBm	45.72
Antenna Gain	-	46.56
HGA Efficiency	%	65%
HGA Size	m	3
Light	m/s	299,792,000
Circuit Loss	-	-2
Pointing Loss	-	-0.17
HGA Pointing Error	deg	0.1
Beamwidth (3 dB)	-	0.83
Total EIRP	dBm	90.11

Path Calculations		
Variables	Units	Values
Space Loss	-	-282.3
Atmosphere Attenutation	-	-0.2
Total Path Loss	-	-282.5

Receiver Calculations			
Variables	Units	Values (70-m)	Values (34-m)
DSN Antenna Gain	-	74.4	68.2
Pointing Loss	-	-0.1	-0.1
Total Receiver Gain	-	74.3	68.1

- **The link budget is the analysis that ‘sizes’ the telecom system, but it is deeply intertwined with the entire EEIS architecture**
  – The following parameters are heavily traded very early in the design process to ensure a robust communication architecture
- **Telecom Subsystem**
  – Frequency Band (eg, L, S, C, X, Ku, K, Ka)
  – Radio/Amplifier Power Input (eg, 1-200 W)
  – Antenna Type (eg, dipole, parabolic) and sizing (eg, 1-3 m dish)
  – Hardware efficiencies & losses (eg, radio, amplifier, circuit, antenna)
  – Operations losses (eg, pointing loss)
- **Space Loss & Atmospheric Attenuation**
  – Signal characteristics (eg, band, polarization, ellipticity, etc.) vs. losses
- **Ground Station & Architecture**
  – Antenna network (eg, DSN) and size (eg, 34-m, 70-m)
  – Pointing Loss (eg, 0.1 deg)
- **Typical Products**
  – Data Rate (bps), margin (dB), and bit error rate (BER)

Antenna Gain



- Antennas are generally characterized as:
  - "Omni" = "low gain" $G \lesssim 3\text{ dB}$
  - Directional = "medium gain" or $5 \le G \le 20\text{ dB}$
    " high gain" $20\text{ dB} \le G$

Beam width (ϕ) = angle between -3dB (half power) points relative to power on boresight axis

Ideal (uniform) beam illuminating 1 deg^2 has a gain of 41,253 (46.2 dB)

(Ideal) $G\phi = 4\pi_{sr} = 41{,}253\text{ deg}^2$

(Real) $G\phi = \eta_a 41{,}253\text{ deg}^2$ where : ϕ = "Field of View" (FOV)

A Typical Antenna Pattern

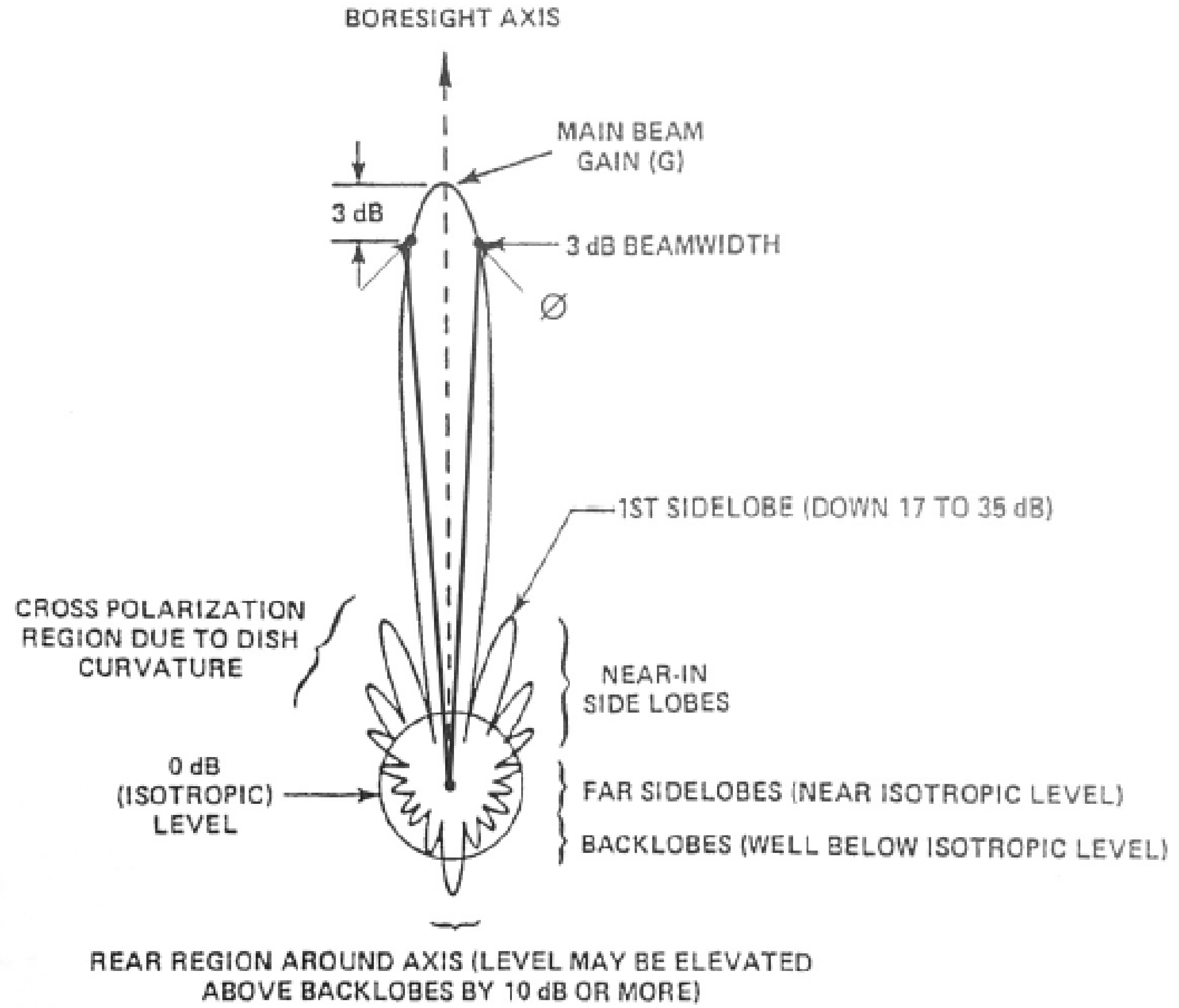


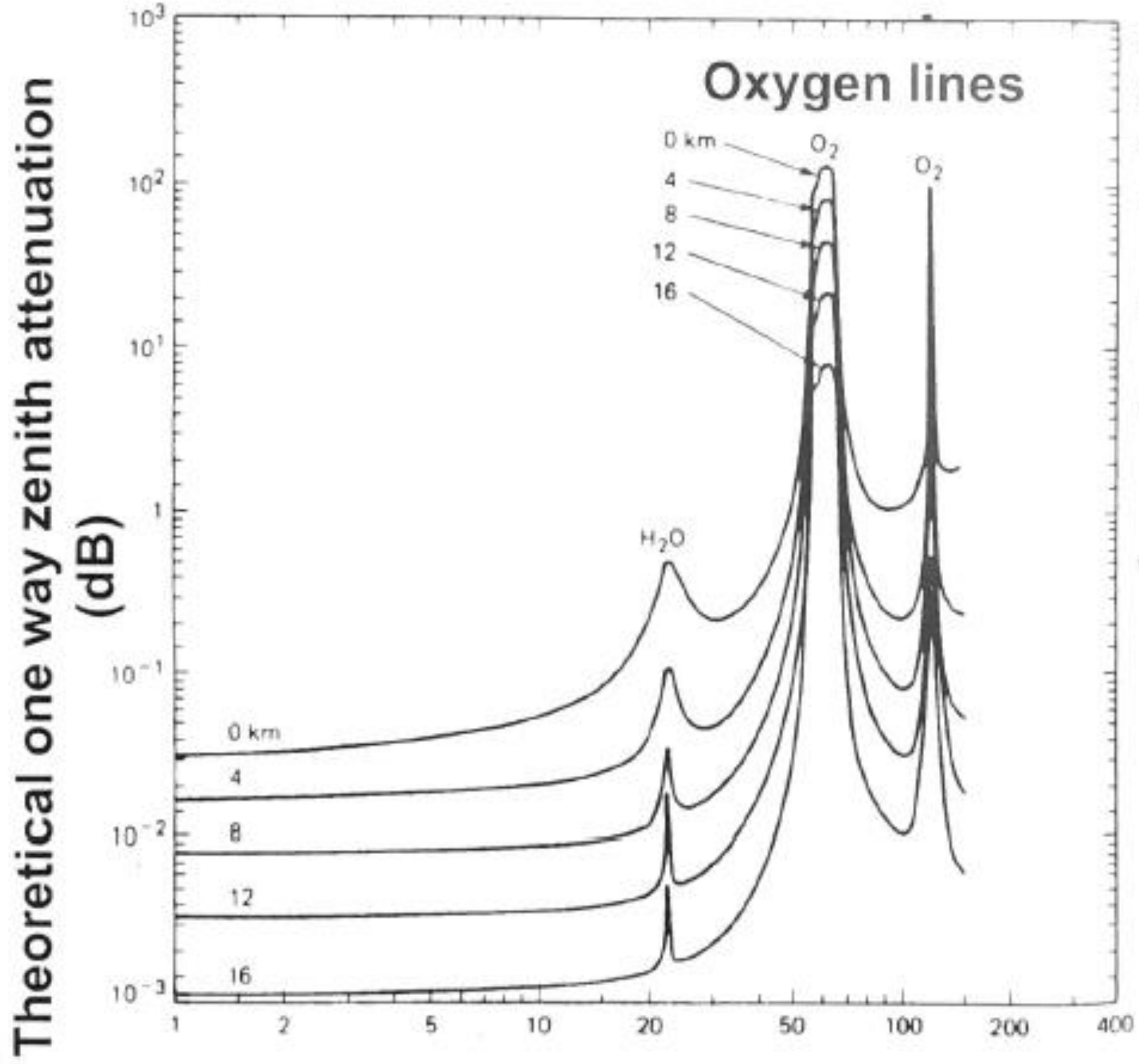
Figure 5

(Griffin & French, p. 439)

Atmospheric Attenuation

- Peaks due to resonance of molecules
- Vibration frequency of Molecule = radiation frequency at peak
- Peaks at 22 and 185 GHz due to water
- Peaks at 60 and 120 GHz due to Oxygen
- Attenuation due to absorption and scattering from suspended particles

Attenuation vs. Frequency

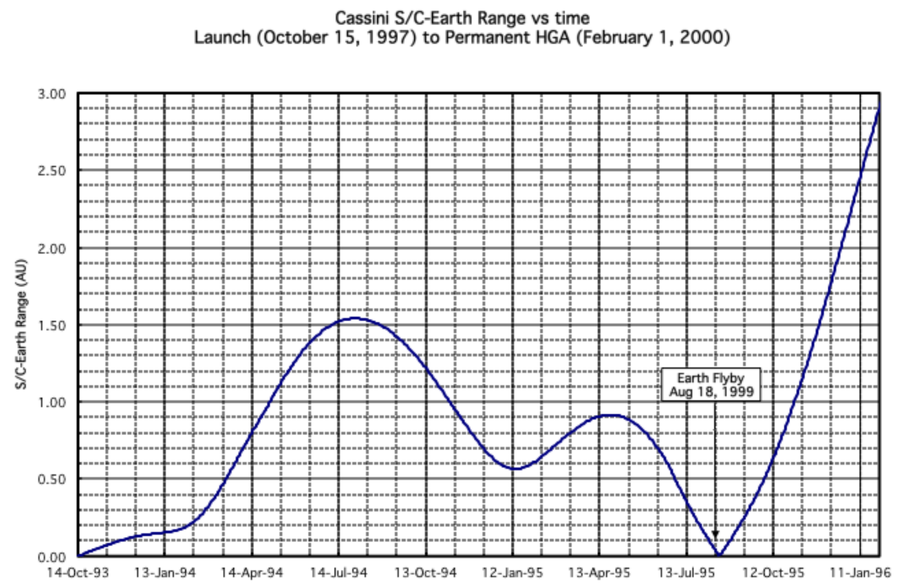


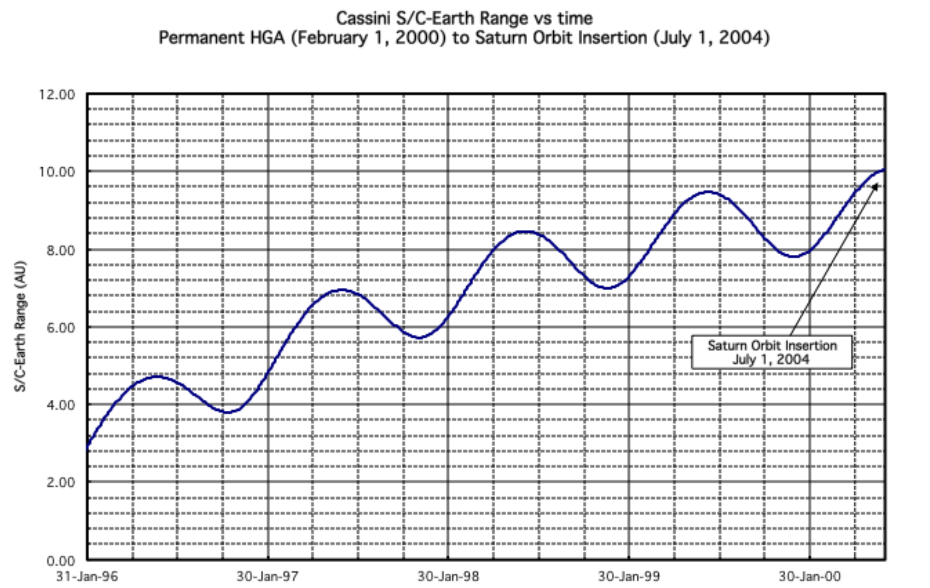
- <0.1 GHz negligible
- >5 GHz significant
- > 20 GHz severe
- Mostly due to water and oxygen
- Decreases with altitude
- Increases with lower terminal elevation angle
- Between peaks are called windows

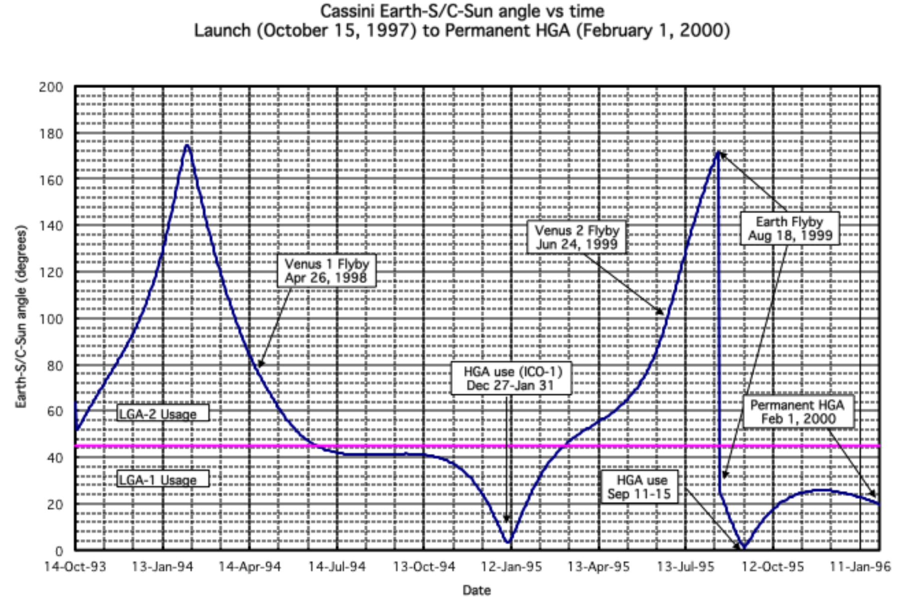
Ref 1

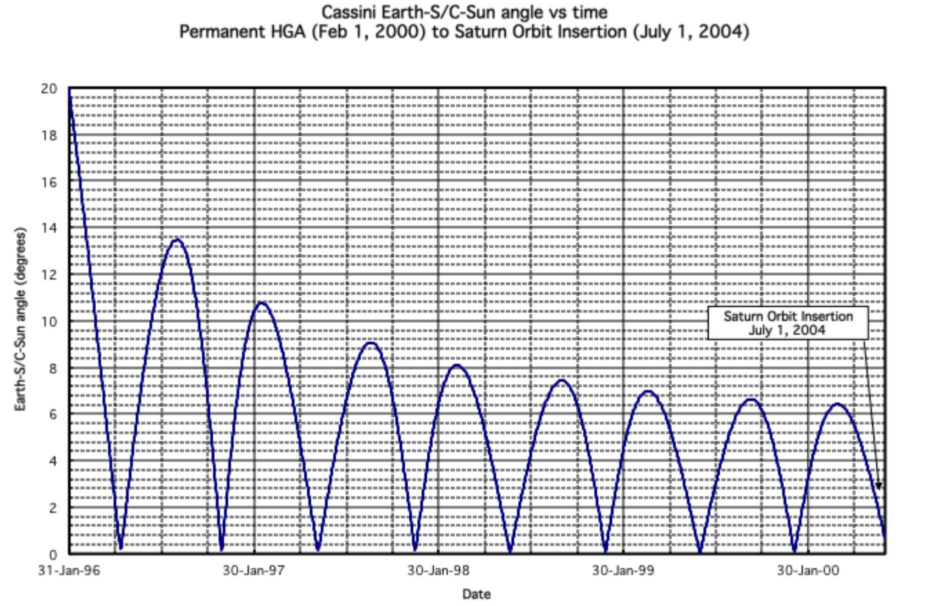
Frequency in GHz

Trajectory Impacts on Telecom







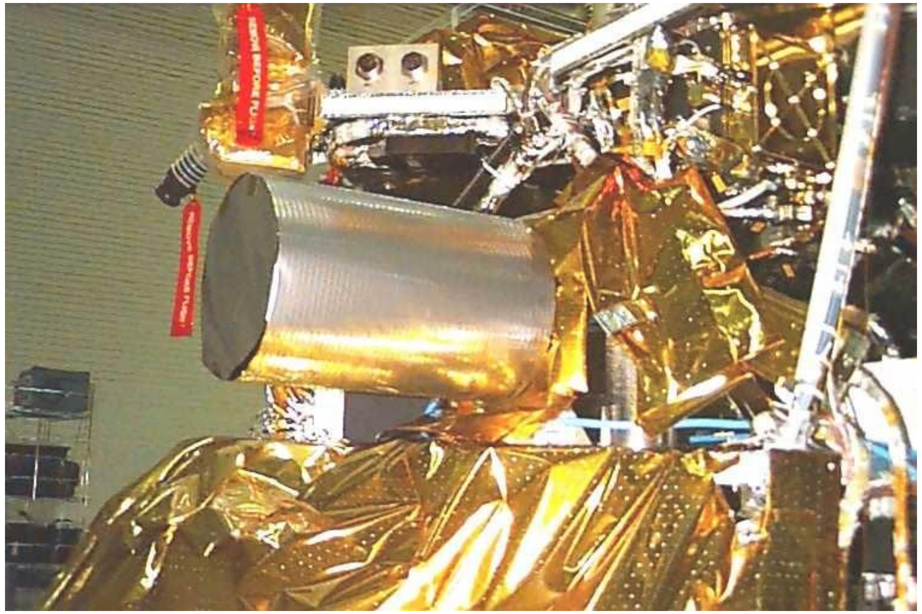


Fields of View (FOV) Impacts…

- Interference with fields of view effect antenna patterns.
- Other items on nadir deck effect pattern. Objects are same order as the wavelength ~ .75m.

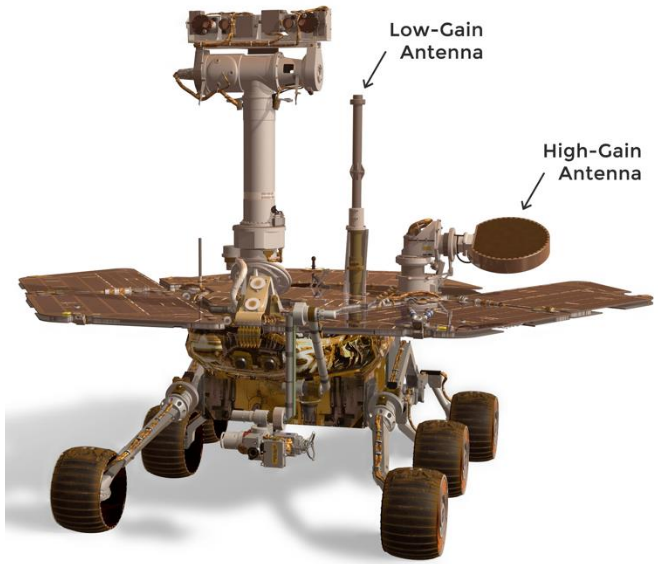
MRO Flight UHF Antenna

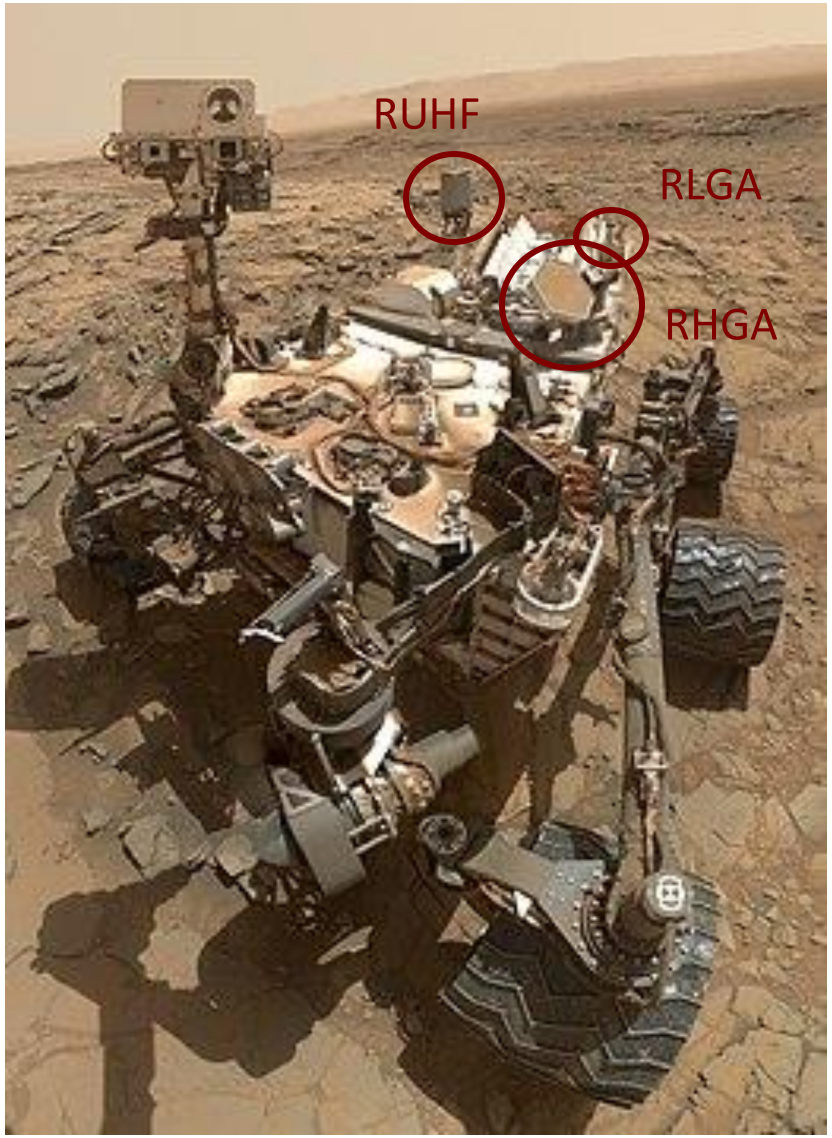
Antenna with radome mounted on MRO nadir deck.



Mars Rovers

- **Typical Curiosity Data Rates**
  - 2 Mbps via RUHF
  - 500 bps via RHGA
  - 10 bps via RLGA



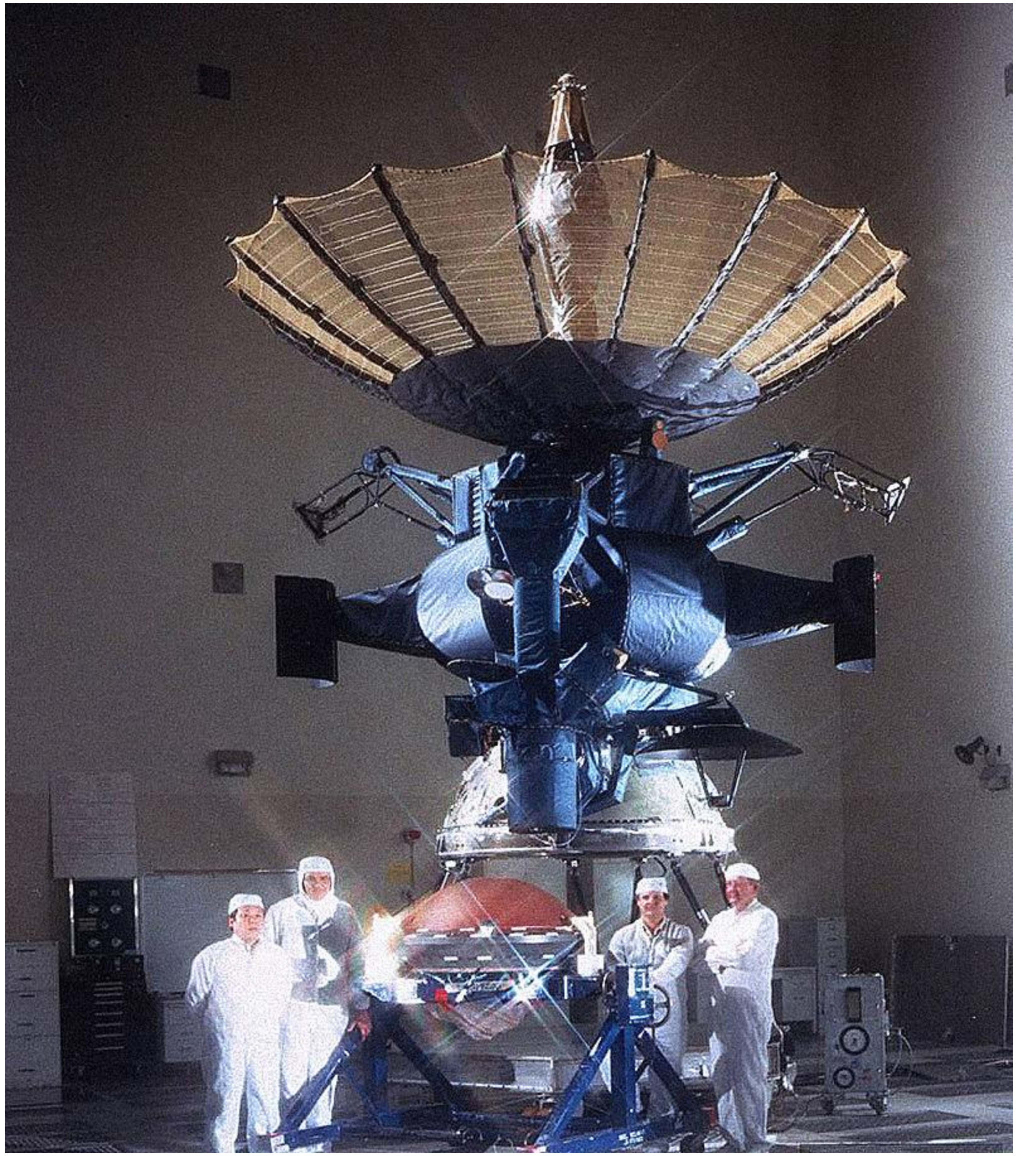


RF Compatibility

- **All subsystems and instruments are required to avoid conducted and radiated emissions and susceptibilities.**
- **RF systems, communications and instruments, are intended to radiate and be susceptible at specific frequencies.**
- **Couplings between each element at each frequency and harmonics must be estimated and controlled.**
- **Filters must be added in the design.**
- **S/C system test must include complete self-compatibility test, including flight configuration, thermal blankets, and payload.**

Case Study

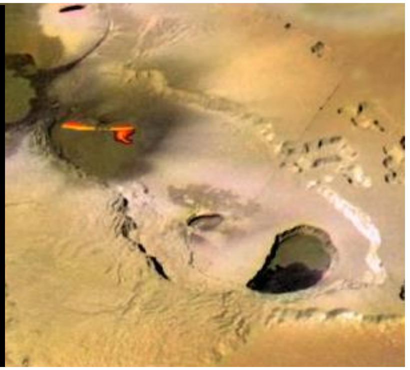
Galileo HGA Failure



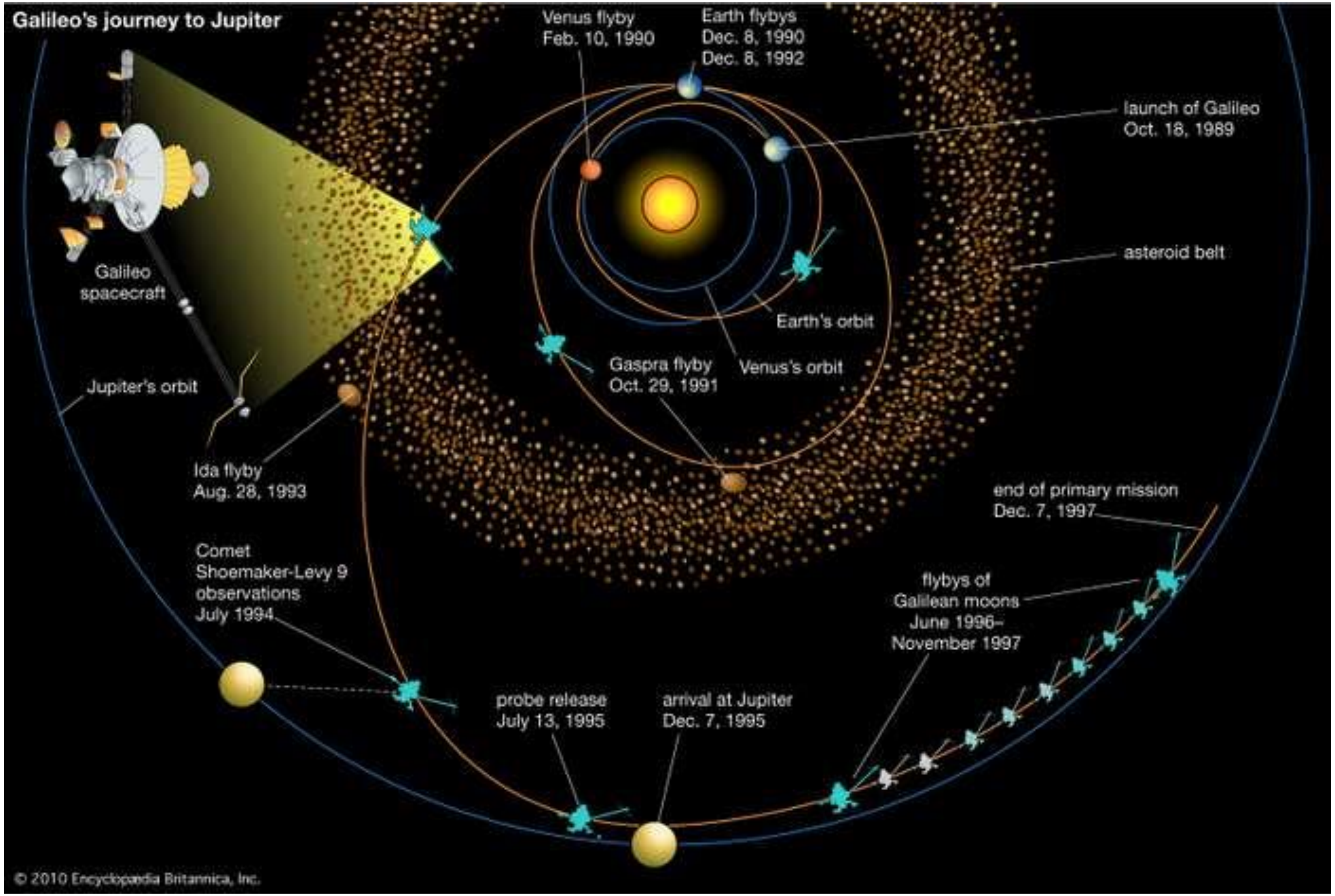
- **Galileo**
  - Launched in 1989 from the Space Shuttle Atlantis
  - Jupiter Orbit Insertion (JOI) in 1995
  - Controlled entry into Jupiter in 2003
- **Scientific Discoveries**
  - Intense radiation environment
  - Liquid ocean under Europa ice
  - Io's active volcanos
  - Ganymead's magnetic field
- **Near Mission Failure….**
  - In 1991, the operations team tried and failed to fully deploy the 4.8-m High Gain Antenna

Europa

Io



Trajectory Overview

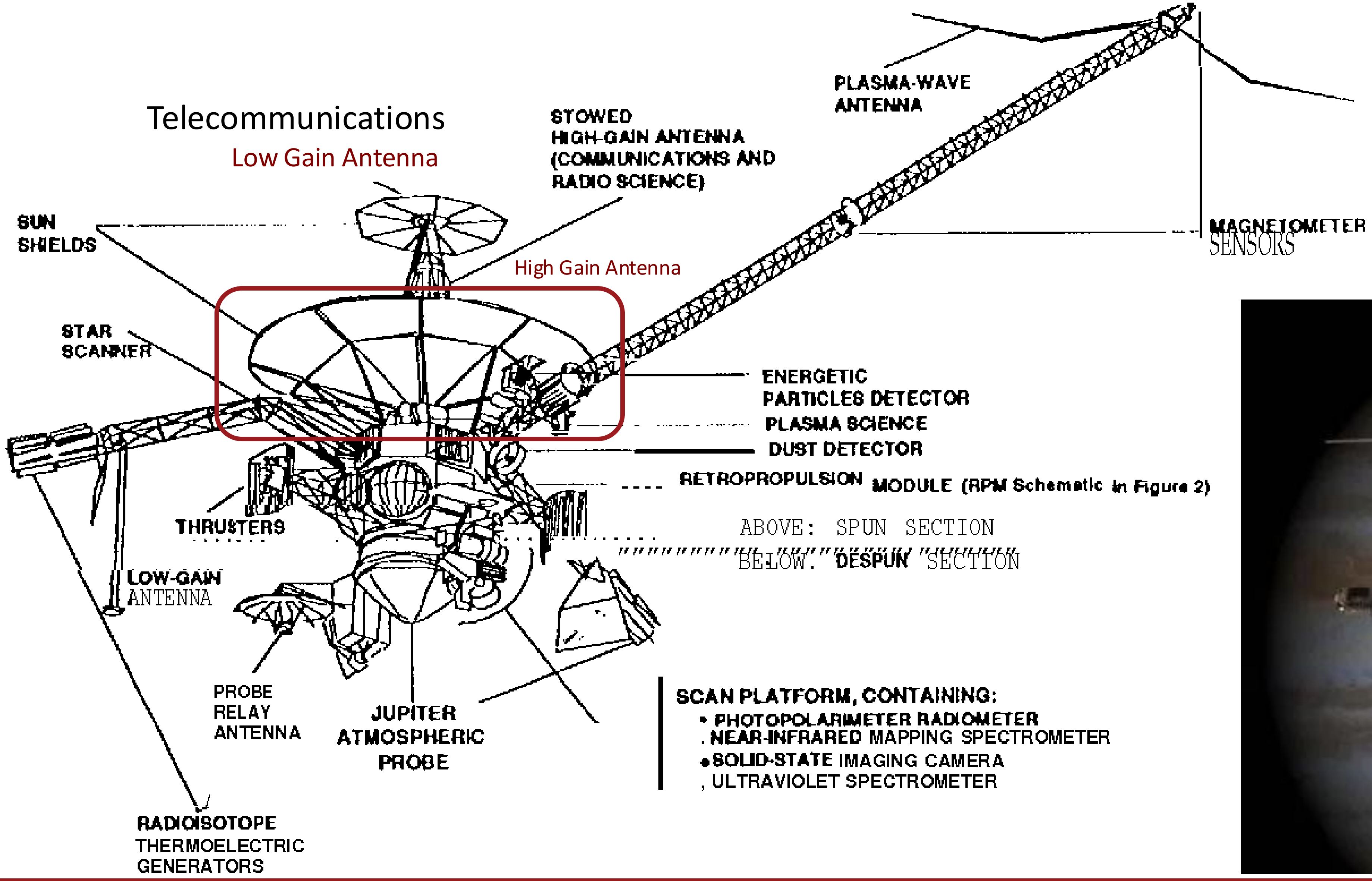


Flight System Overview

Telecommunications

Low Gain Antenna

High Gain Antenna

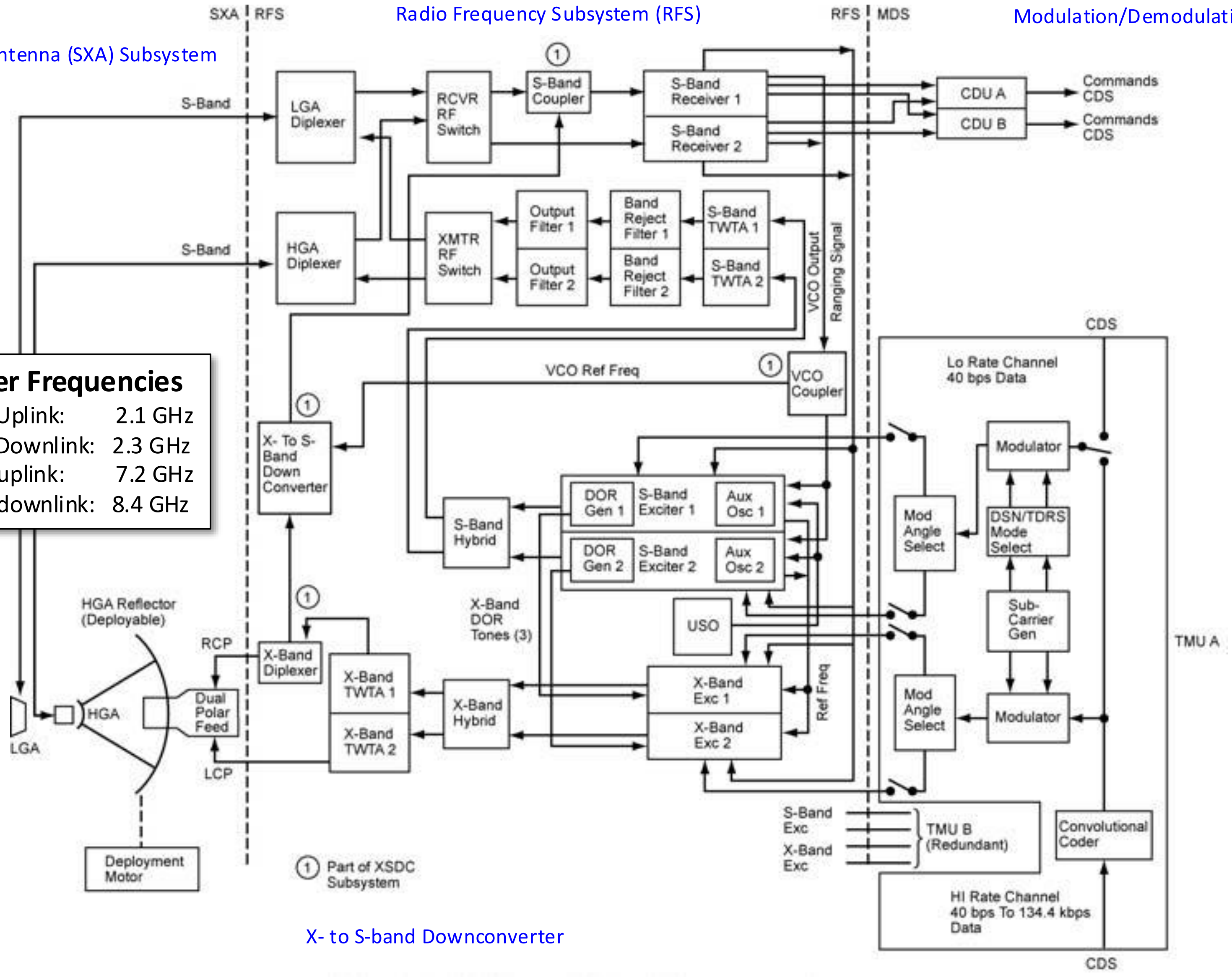


Telecommunications Overview

S-/X-band Antenna (SXA) Subsystem

Radio Frequency Subsystem (RFS)

Modulation/Demodulation Subsystem (MDS)



Carrier Frequencies

S-band Uplink:	2.1 GHz
S-band Downlink:	2.3 GHz
X-band uplink:	7.2 GHz
X-band downlink:	8.4 GHz

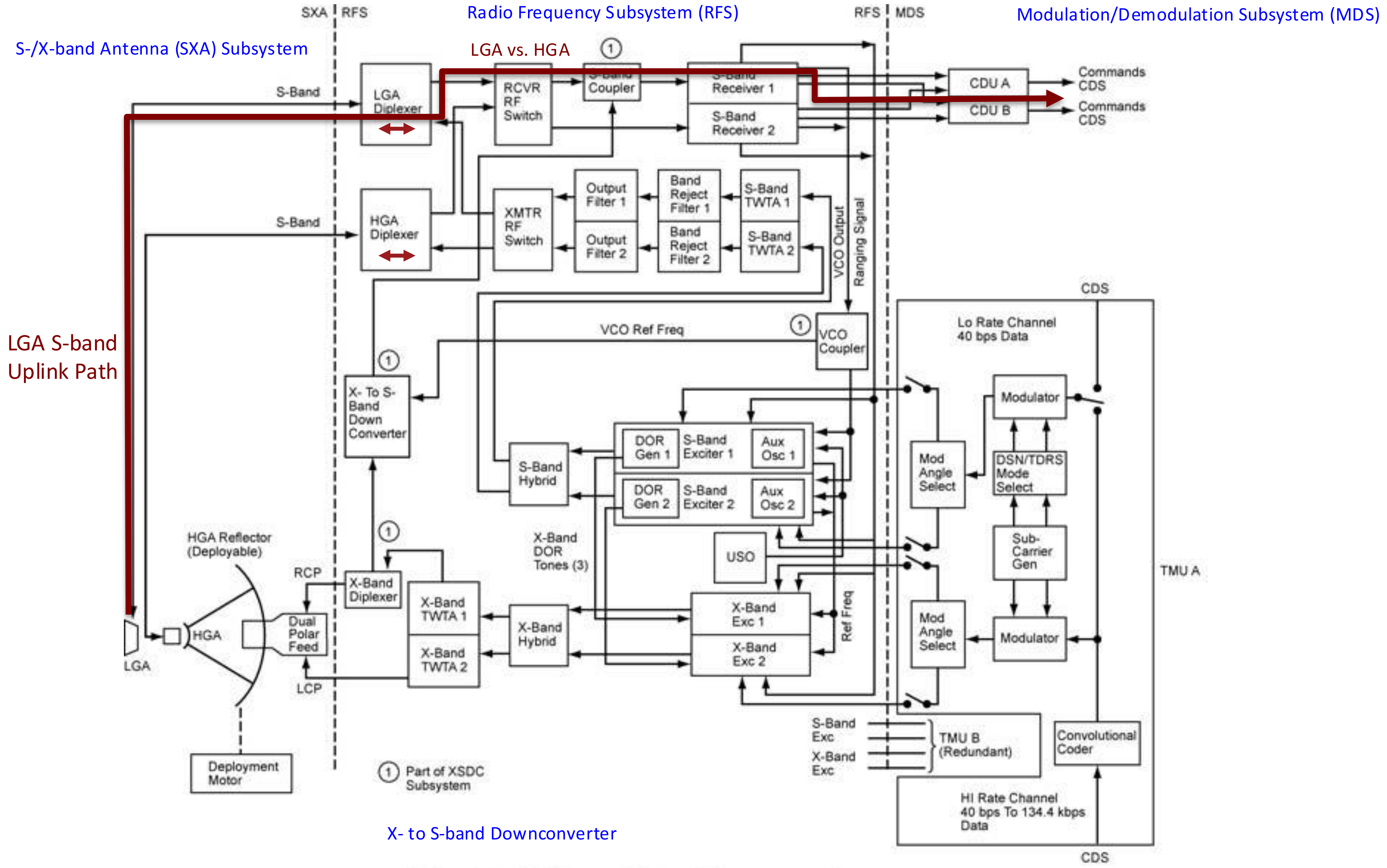
X- to S-band Downconverter

Component Terminology

- **Diplexer:** Routes two signals into one path based on frequency (typically reciprocal)
- **Switch:** Allows the spacecraft to switch between two RF paths (eg, LGA vs. HGA)
- **Filters:** Allow/reject certain frequencies. There are four types: high-pass, low-pass, band-pass, and band-reject
- **TWTA:** Traveling Wave Tube Amplifier is a specialized vacuum tube that is uses additional power (in W) to amplify RF signals
- **Down-converter:** Converts RF signals to a lower frequency range.
- **Hybrid Coupler:** Splits power evenly (3 dB) between two ports
- **Exciter:** Generates the modulated waveform that can be radiated to the antenna.
- **RF Oscillator:** Generates the carrier signal that is used for spacecraft communications. An ultra-stable oscillator (USO) is a highly stable version generally used for science.
- **Modulator:** Converts signals from digital to RF
- **Convolutional Code:** Type of error-correcting encoding that is used to increase the robustness of spacecraft communications. Notes: (1)Andrew Viterbi proposed the Viterbi algorithm to decode convolutionally encoded data. (2) Convolutional codes are being replaced by Turbo codes.

(These definitions are intended to be more readable, but slightly less accurate.)

Telecommunications Overview



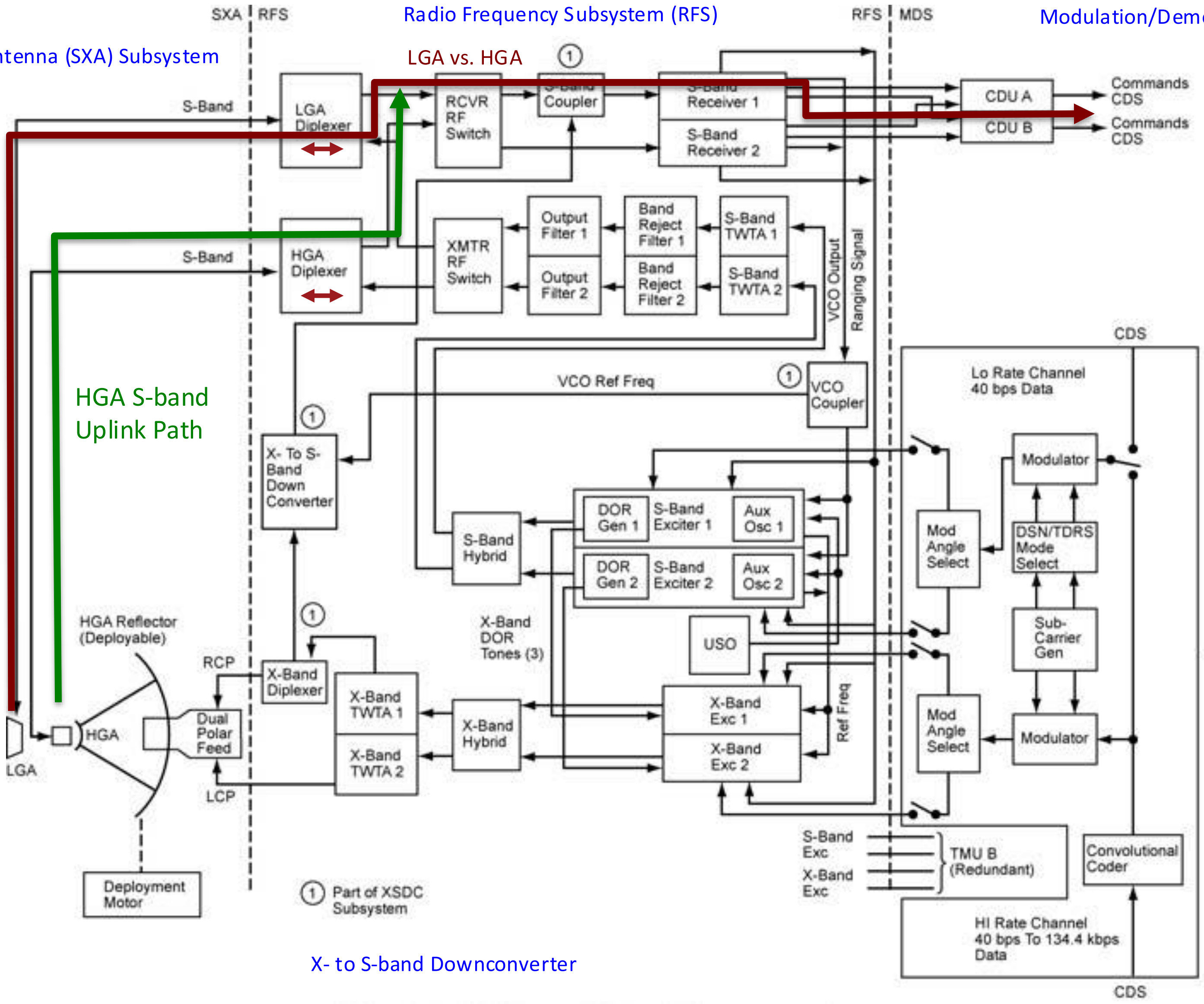
Telecommunications Overview

Radio Frequency Subsystem (RFS)

Modulation/Demodulation Subsystem (MDS)

S-/X-band Antenna (SXA) Subsystem

LGA vs. HGA



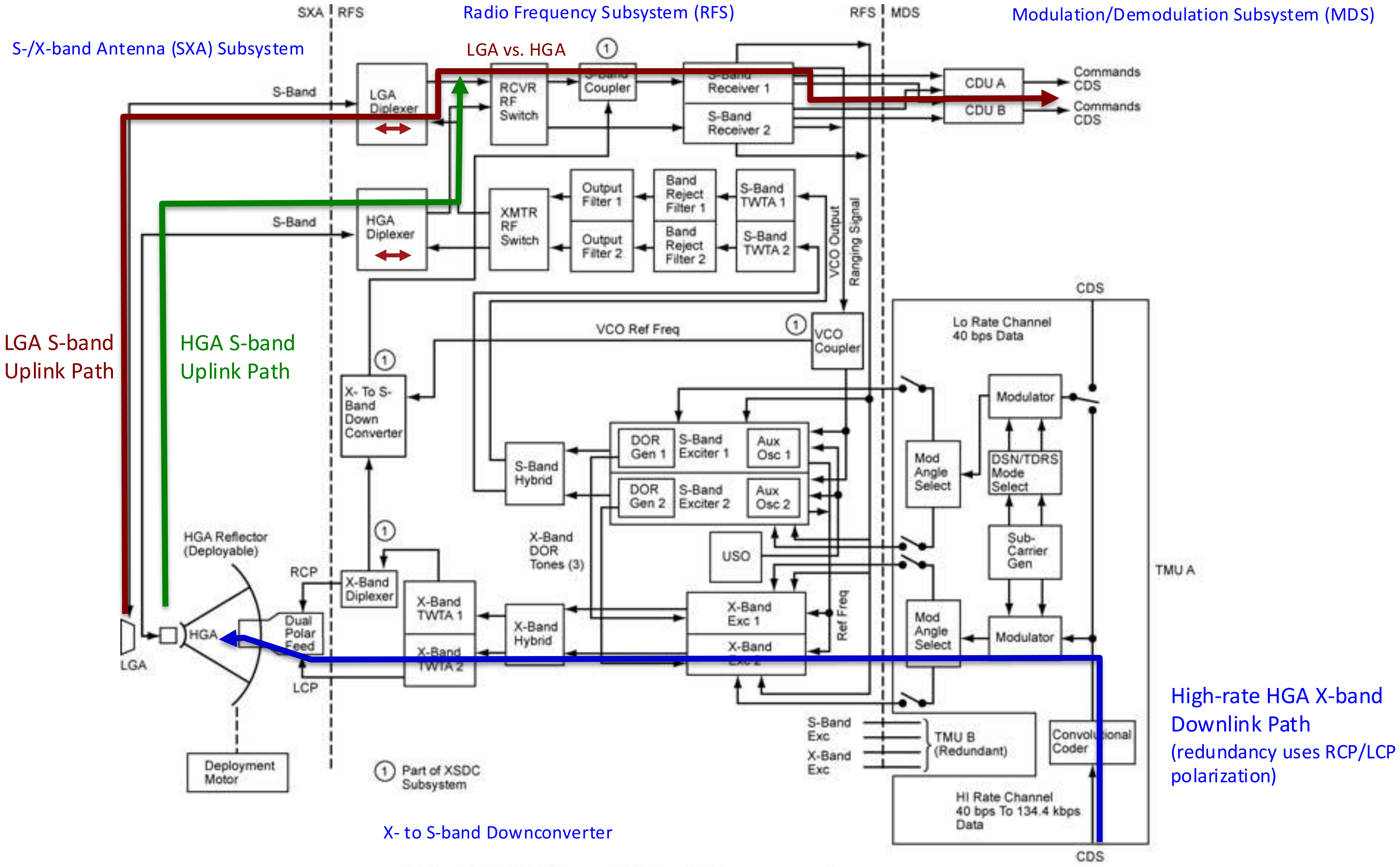
Low-rate LGA S-band Downlink Path

LGA S-band Uplink Path

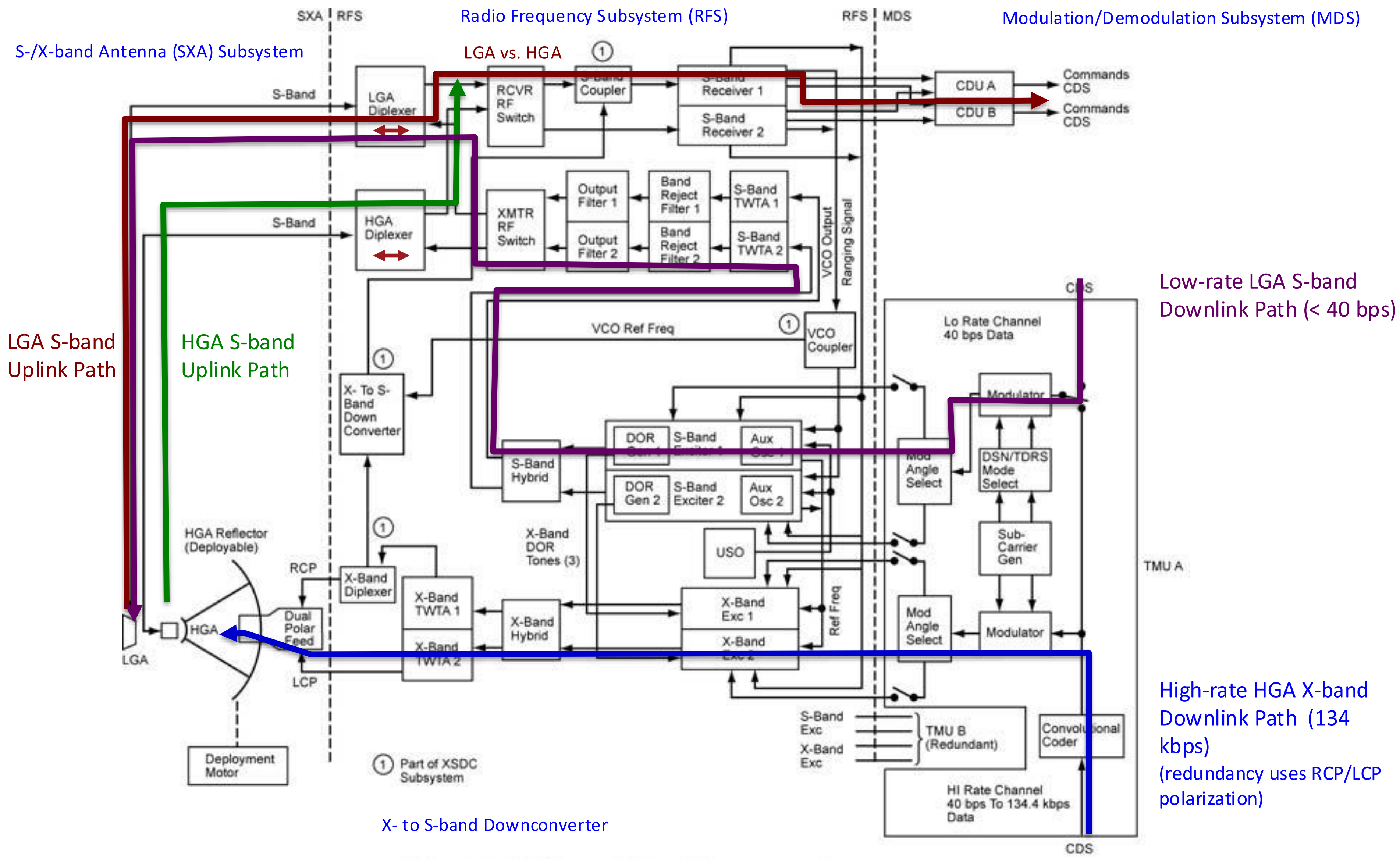
HGA S-band Uplink Path

X- to S-band Downconverter

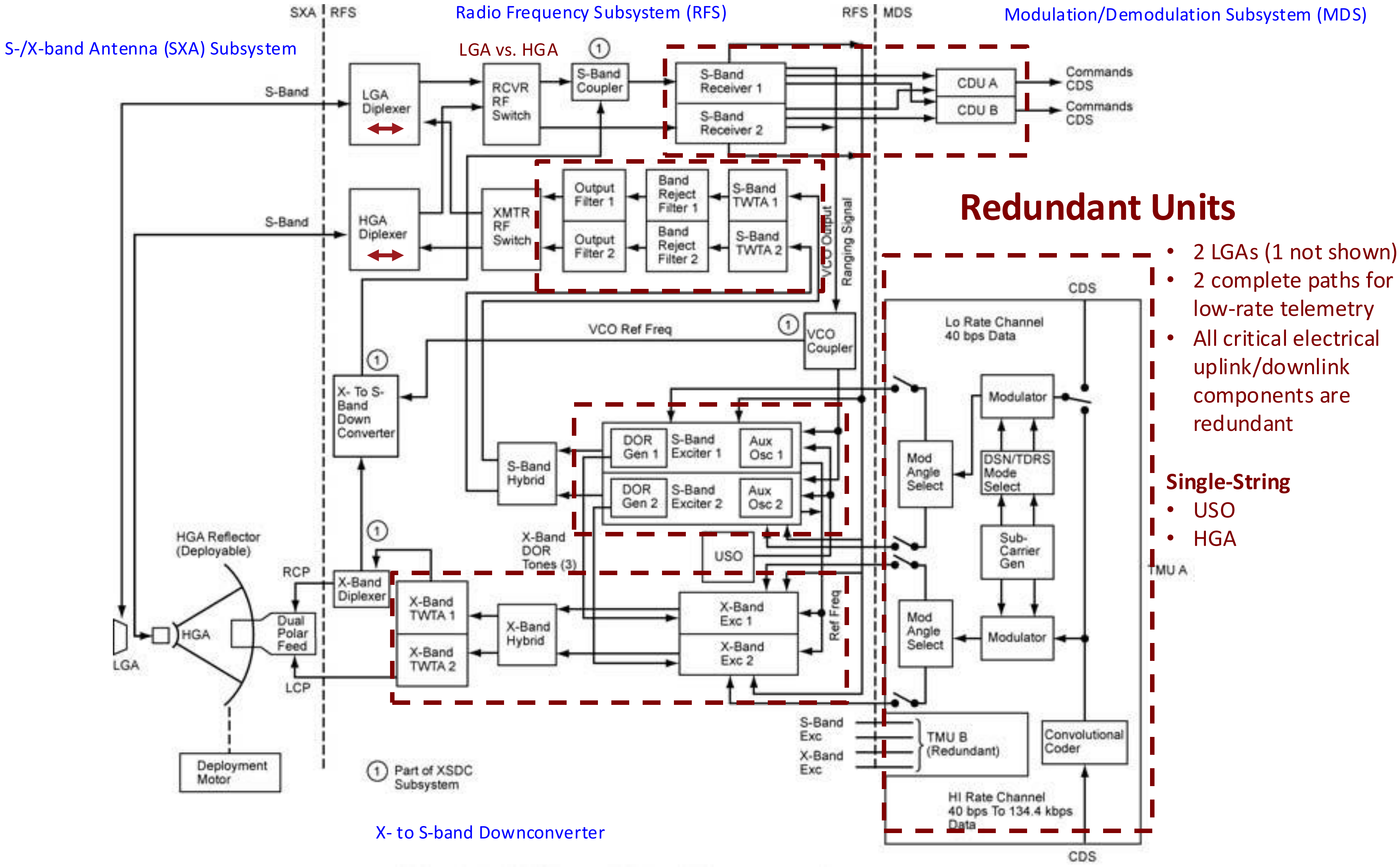
Telecommunications Overview



Telecommunications Overview



Redundancy



Redundant Units

- 2 LGAs (1 not shown)
- 2 complete paths for low-rate telemetry
- All critical electrical uplink/downlink components are redundant

Single-String

- USO
- HGA

Since Galileo…

Small Deep Space Transponder (SDST)

Solid State Power Amplifier (SSPA)

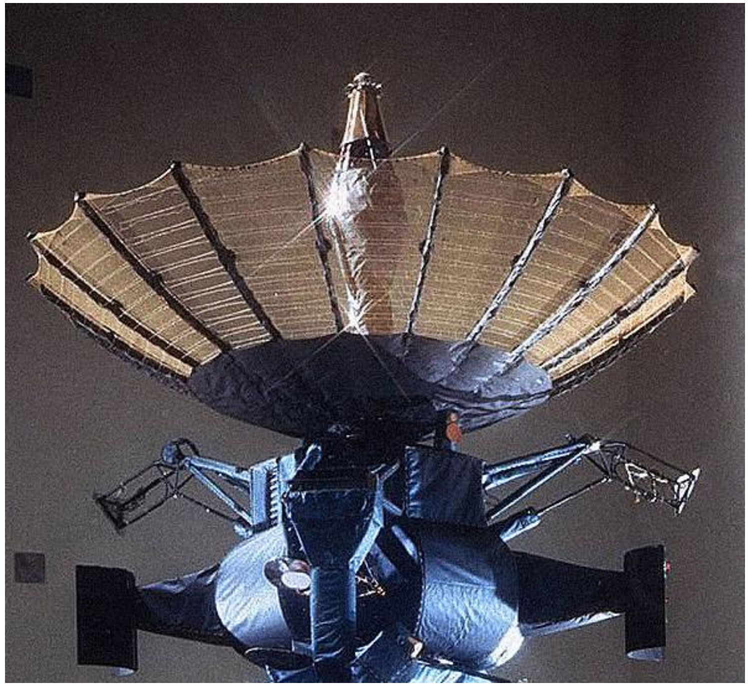
- **SDST**
  - Unifies the following functions
    - Receiver
    - Command detector
    - Telemetry modulator
    - Exciters
  - Note that encoding and compression is typically performed by the C&DH subsystem
- **SSPA or TWTA**
  - Solid State vs. Traveling Tube
  - Uses power to amplify the signal
  - Often range from 25 W to 100 W
- **Result**
  - The above components greatly simplify modern telecom systems
  - Typically, just combine these with
    - Antennas
    - Diplexers & Switches (waveguide/coaxial)

Initially Anomaly…

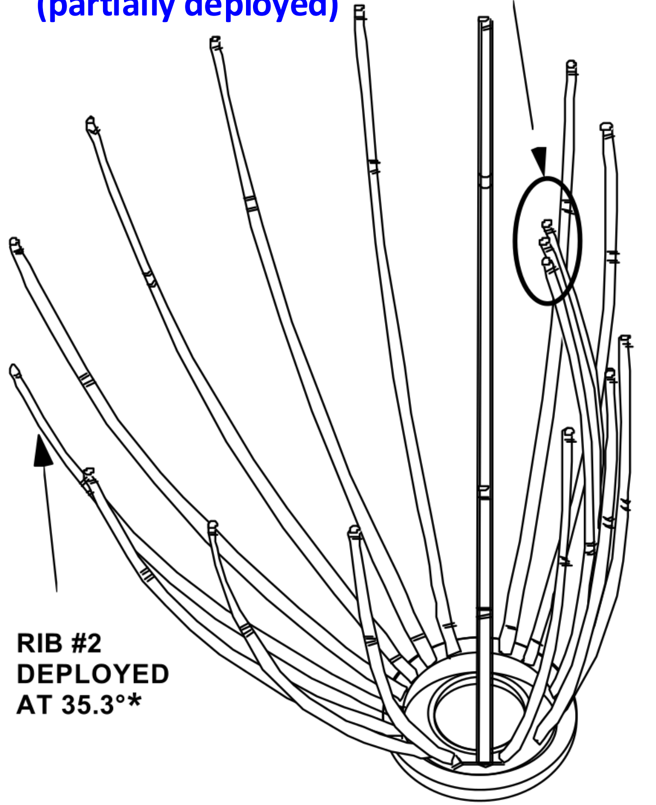
- **In 1991, after the Venus flyby and beyond 1 AU, the team executed a sequence to deploy the high gain antenna**
- **While the telemetry showed that the motors operated at higher-than-expected power levels for 8 minutes, they were unable to deploy the antenna.**
- **3 ribs were 'stuck'**

(Can occur when there is insufficient lubricant)

Expected State (fully deployed)



Actual State (partially deployed)



* AFTER HAMMERING RIB #2 NOW AT 43°

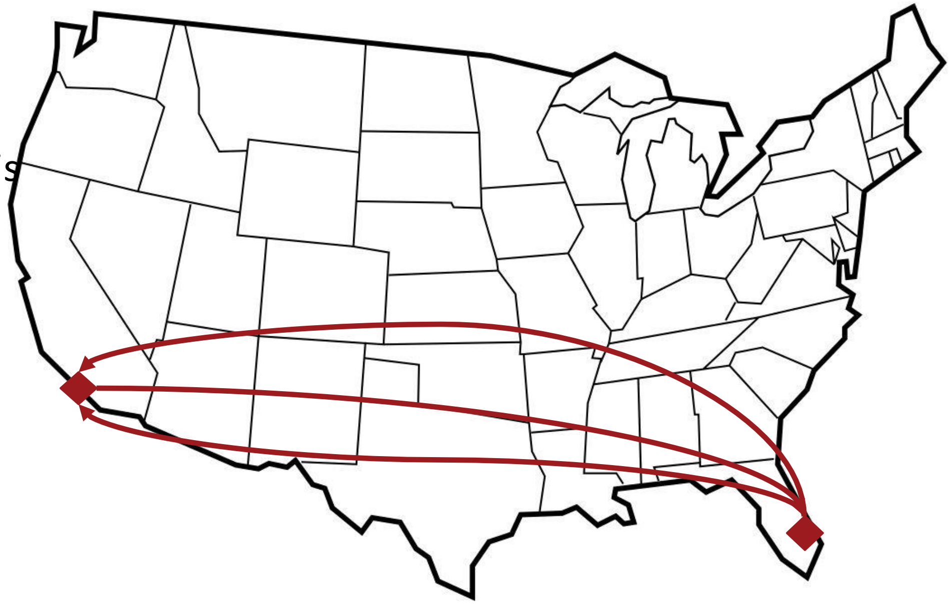
Why?

- **Probable root cause…**
  - Loss of lubricant occurred due to the vibration that the antenna experienced on the truck as it made multiple trips to/from KSC
    - Failed ribs were closest to the flatbed trailers that carried Galileo
    - The lubricants were not checked or replaced prior to launch

Timeline

- Dec, 1985- Originally shipped from JPL to KSC
- Jan, 1986- Loss of Space Shuttle Challenger
- Late 1986- Returned to JPL
- Mid, 1989- Shipped to KSC
- Oct, 1989- Launched on Space Shuttle Atlantis
- Early 1991- HGA deployment failed

(Picture from Phoenix S/C)



Attempts to Unstick HGA

- **Attempts to Unstick the HGA**
  - Changing orientation towards/away from sun to change temperature
  - Repeatedly turning on/off the motors would double the torque (13,000 attempts)
  - Inducing a maximum spin rate of 10 rpm & hammering the motors
  - Final attempts coincided with release of the probe, probe deflection maneuver, and JOI
  - None were successful…
  - At Jupiter, downlink rate would be 10 bps, which would effectively end the mission

Transmission Improvements & Results

- **From 1993 to 1996, extensive new flight and ground software was developed and DSN stations were enhanced to increase the transmission rate**

- **Improvements**
  1. Increased S-band signal to noise ratio (antenna arrays & ultra-low noise receivers)
  2. Improved efficiency of radio modulation
  3. Improved channel codes (ie, reduce Eb/No) so to minimize BER
  4. Aggressively apply data compression techniques

- **Result**
  – 1-3 improved the downlink rate from 10 bps to 160 bps
  – 4 improve the rate to 1,000 bps
  – These improvements allowed Galileo to meet 70% of its science goals

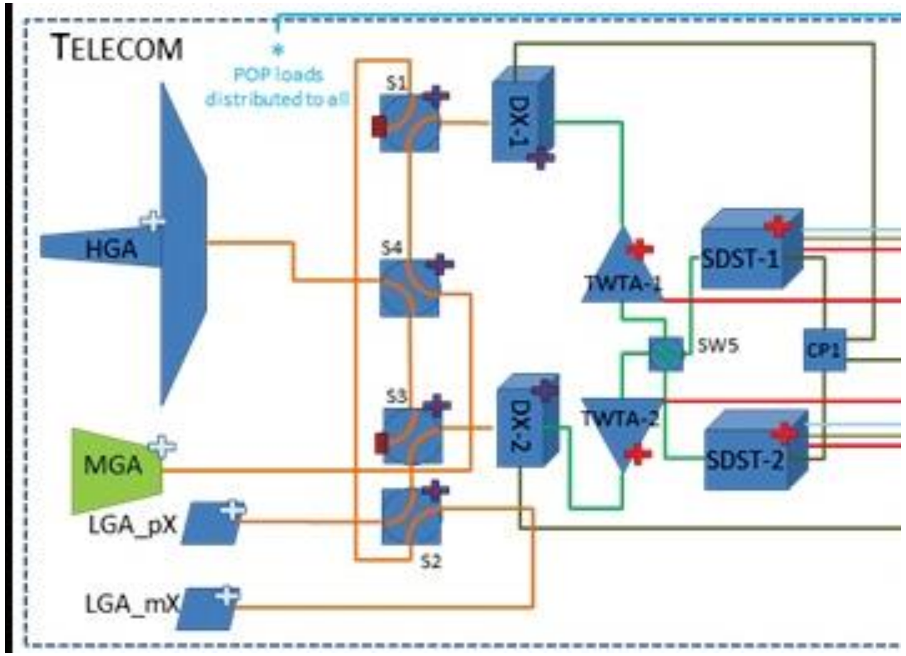
Beyond Galileo, these changes both significantly improved NASA deep space communications and changed how the science community viewed data compression.

Telecom Hardware Estimate

- **Primary Components (part of link budget)**
  - Radio / Transponder, find lowest mass, power, & cost system that meets requirements
  - Power Amplifier
    - Use a TWTA or SSPA (often depends on specific design and/or heritage)
    - Otherwise, just based on lowest mass/cost/power system that provides desired RF power
  - Antennas
    - High Gain antenna for primary data uplink/downlink
    - Medium gain antenna when high gain not required due to lower rate or nearer range
    - Low Gain antenna for fault scenarios (~360-deg coverage)

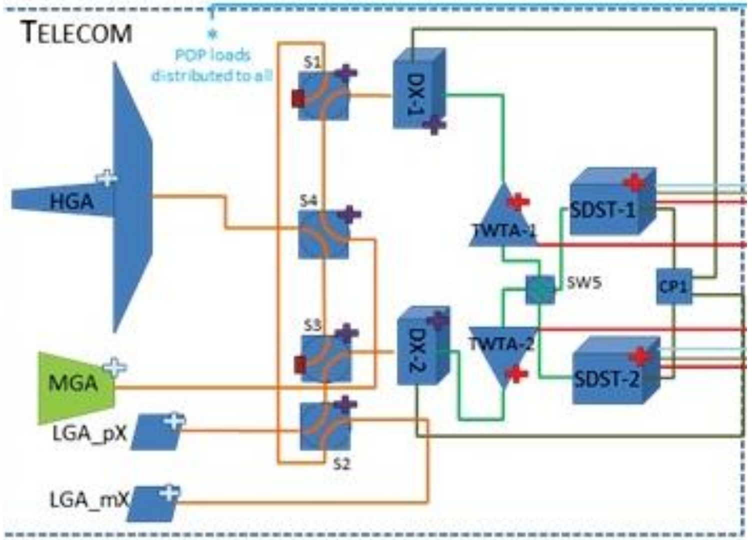
- **Additional Hardware**
  - Diplexers
    - Routes different frequencies
    - Uplink/downlink, S- vs. X-band
    - 0.7 kg each (ballpark)
  - Waveguide Transfer Switches
    - Configures between different antennas
    - Depends on design, but 1 per 1-2 antennas
    - 1 kg each (ballpark)
  - Coaxial Cable, Filters, etc.
    - ~5% of telecom mass

OSIRIS-Rex Telecom Subsystem



Telecom Design Steps

- **Review & Understand Design Information**
  - Mission Description and/or Concept of Operations
  - System and Subsystem Requirements
    - ConOps, mission geometry, Earth-S/C range
    - Payload & engineering data requirements
    - Fault scenarios



- **Create a Preliminary Design**
  - Identify the driving uplink/downlink scenarios
  - Create link budgets for each
    - Optimize with respect to band, antenna types & sizes, RF power, ground station assumptions, etc.
  - Create block diagram
  - Create component mass list

- **Review & Iterate (w/broader team)**
  - Revisit other options & trades

QUESTIONS

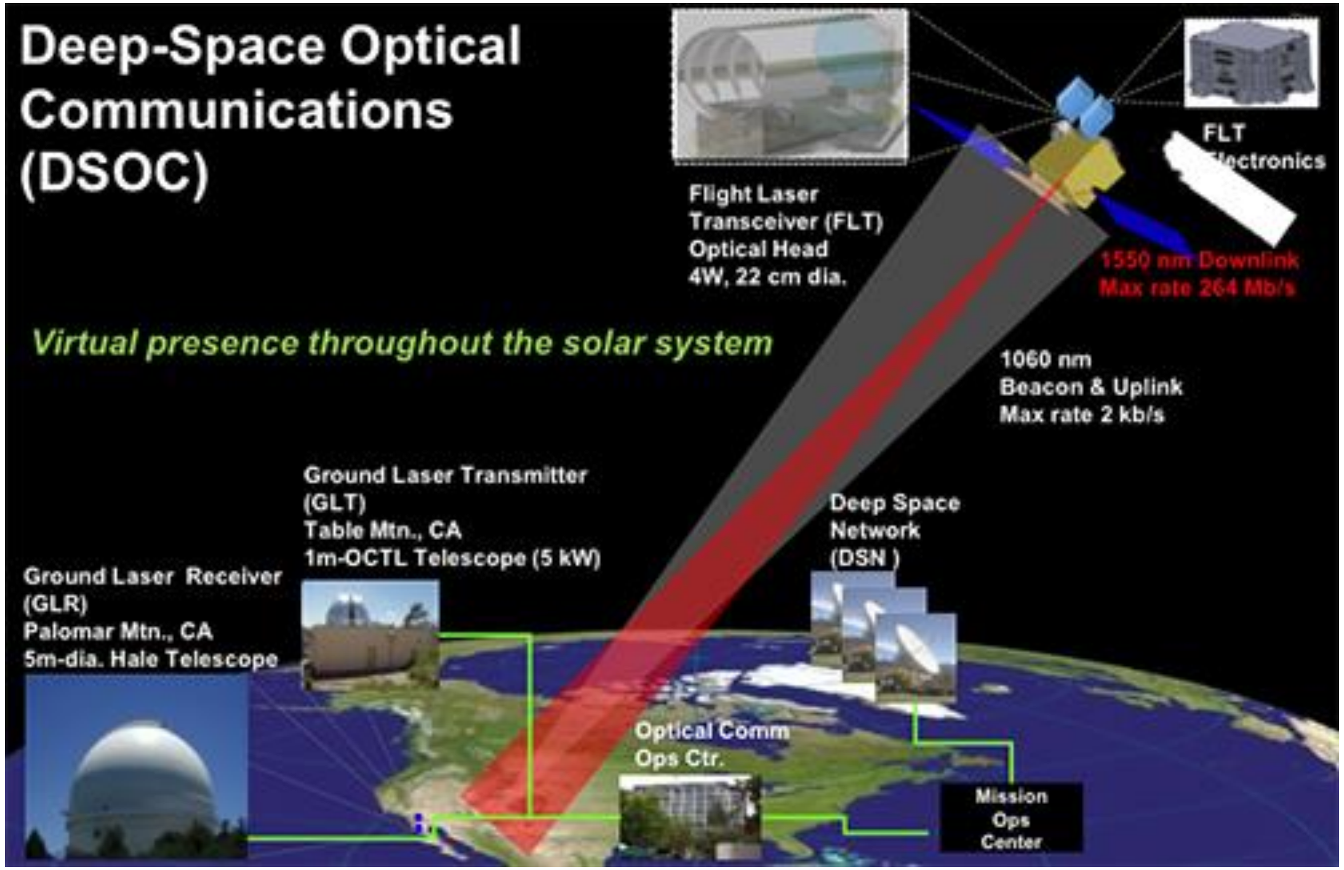
Questions from Last Week

- **Telecom is a critical subsystem, but are there any spacecraft that might not need it or where it's less important?**
  - Thoughts from the class?

- **Are there any new innovations/improvements being made to the way s/c communicate either with other s/c or ground stations?**
  - Most significant is optical communication… (next chart)
  - Additionally, there is a continuing drive towards higher bandwidths and higher power systems, which have resulted in an increasing amount of data

Optical Communication

- **Deep Space Optical Communications (DSOC) is a laser space communication system in development meant to improve communications performance 10 to 100 times over the current radio frequency technology without incurring increases in mass, volume or power.**
- **Mass 29 kg, power < 100 W, 0.2-200 Mbps**



Questions from Class (3 of 3)

- **3 dB Beamwidth is the angle at which the signal is a half-power**
  - For example: dB = 10 Log_{10} (unitless factor)

    3 dB = 10 Log_{10} (0.5)

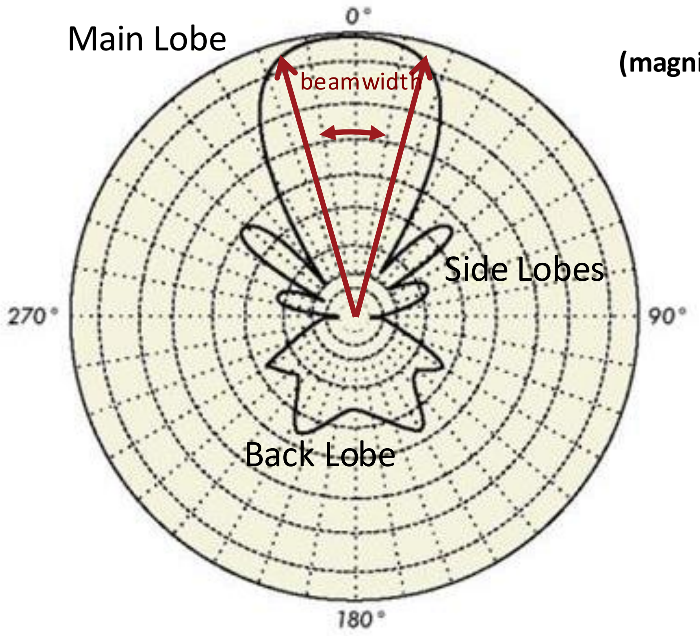
- **For a parabolic antenna**

$$BW = \frac{70\lambda}{d}$$

where BW = antenna beamwidth (deg)

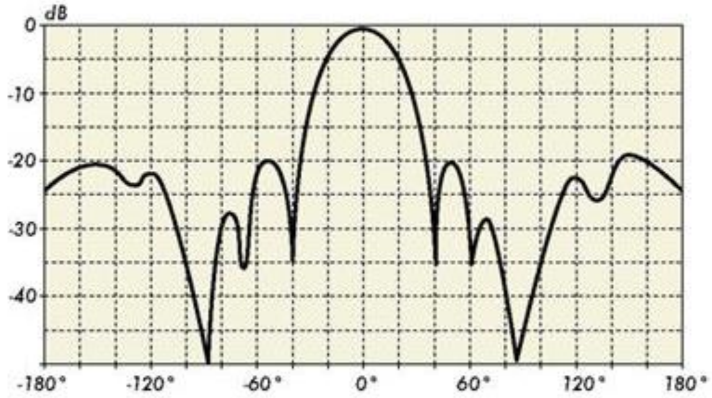
λ = wavelength (m)

d = antenna diameter (m)



Antenna Gain Patterns

(magnitudes are in dB; ie, logarithmic scales)



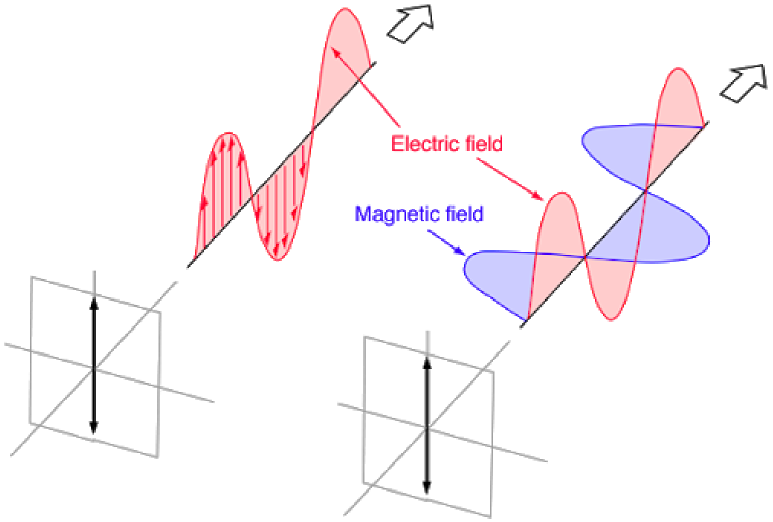
Useful Source

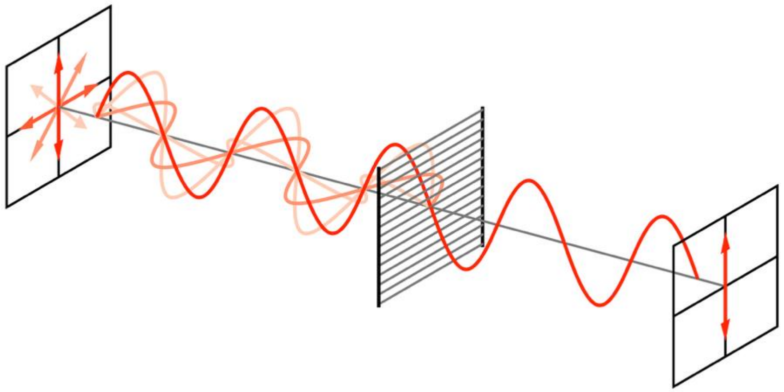
https://www.phys.hawaii.edu/~anita/new/papers/militaryHandbook/antennas.pdf

Questions from Class (4 of 9)

- **"When looking at a telecommunications block diagram, are the block always pieces of hardware?"**
  - For telecom & most subsystems, yes. For, GN&C and Software, no.
  - Note that block diagrams can be used in different ways (hardware, software, functional, organizational, etc.) Important to choose the right "architectural view or perspective" depending on the problem.

- **"If switching between RCP and LCP may cause problems with receiving the signal on the ground, why do it?"**
  - In general, using polarization minimizes interference from other sources and expands the available bandwidth for communications.
  - I believe it's also used, on occasion, when uplinking/downlinking at the same time

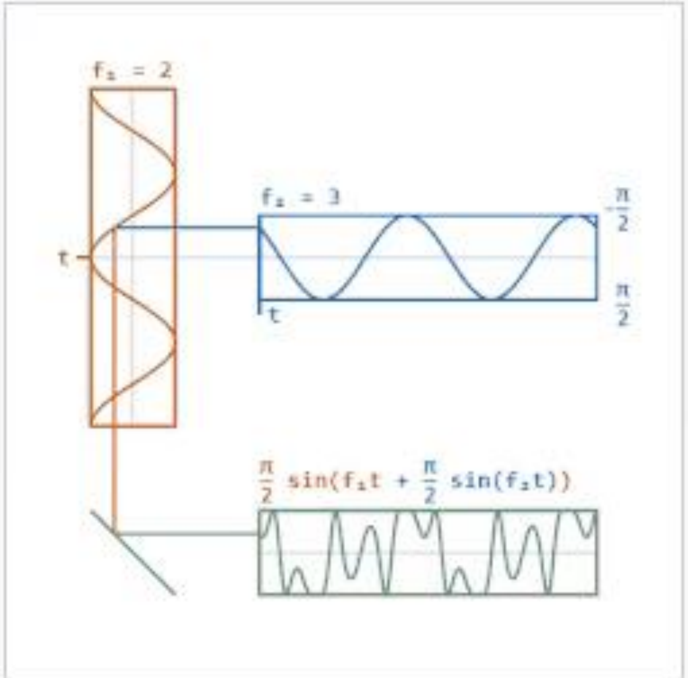




Questions from Class (5 of 9)

- **"Can an antennas receive and send out information at the same time?"**
  – Yes. I believe there is a slight loss when simultaneously transmitting/receiving data
  – Note that the diplexer is responsible for routing the appropriate signals (band & direction)

- **"How is data modulation done and why is it important when downlinking?"**
  – Transmitting data via radio waves requires us to translate real data (ie, bits) **into the RF signal**. This is what modulation is (versus an electrical line where digital signals can be transmitted directly).



The modulating wave (blue) is modulating the carrier wave (red), resulting the PM signal (green). g(t) = π/2 * sin(2*2πt+π/2*sin(3*2πt))

Questions from Class (8 of 9)

- **“For clarification, does the [Telecom] subsystem receive RF signals from the uplink path, transform them into digital signals and then send these digital signals to the C&DH subsystem?”**
  - Yes – this is exactly how it works.
- **“And then when the [Telecom] subsystem is sending signals on the downlink path back to earth, do these signals originally come from the C&DH subsystem?”**
  - Yes – they are either collected and/or generated by FSW, encoded, and then sent to telecom